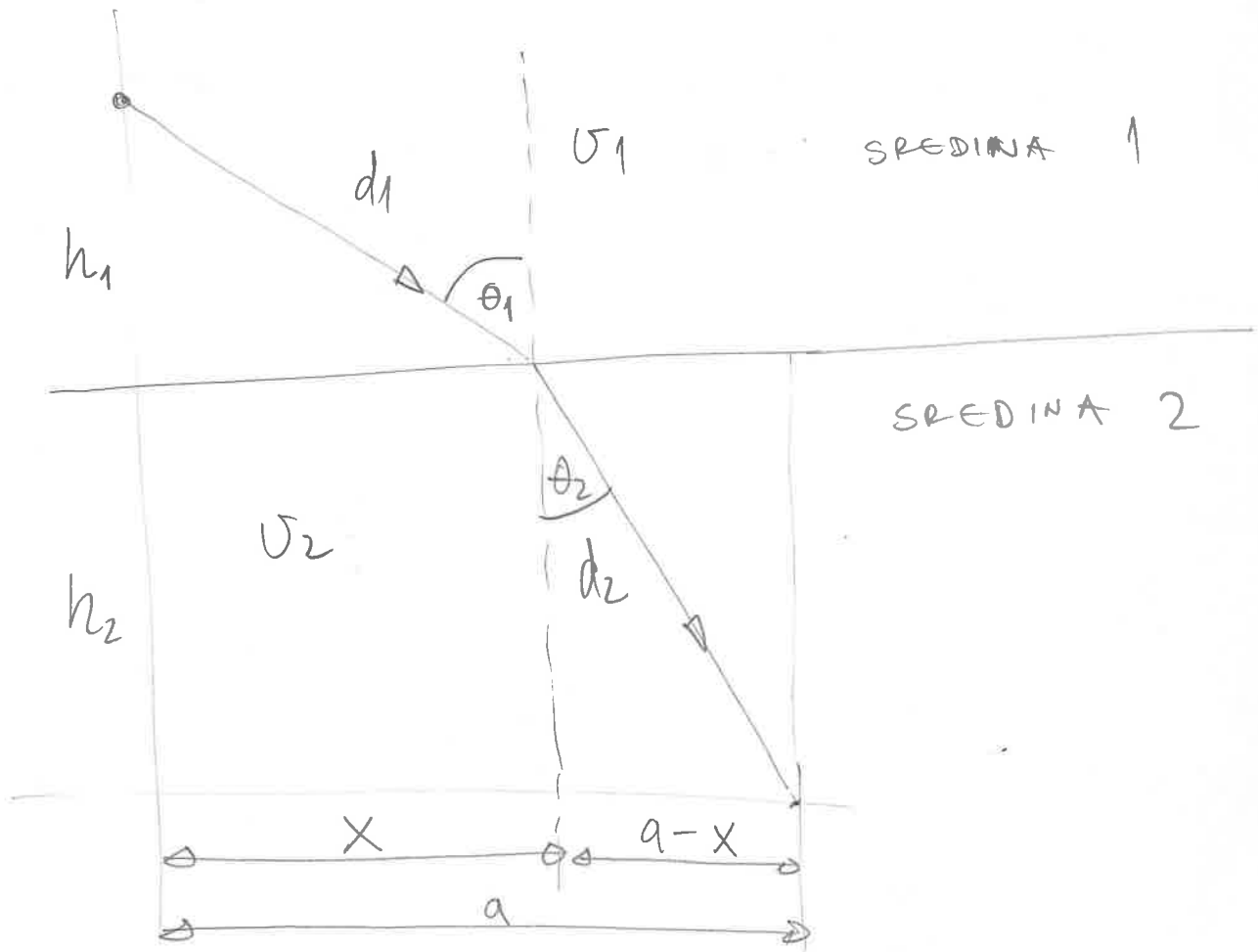


FERMAOV PRINCIP

SVETLOST SE KREĆE PUTANOM KOJA IMA
NAJKRATČE VREME



$$t = \frac{d_1}{v_1} + \frac{d_2}{v_2} \quad \text{UKUPNO VREME}$$

$$d_1 = \sqrt{x^2 + h_1^2}$$

$$d_2 = \sqrt{(a-x)^2 + h_2^2}$$

$$t = \frac{\sqrt{x^2 + h_1^2}}{v_1} + \frac{\sqrt{(a-x)^2 + h_2^2}}{v_2}$$

$$\frac{dt}{dx} = \frac{x}{\sqrt{x^2 + h_1^2} \cdot v_1} - \frac{(a-x)}{\sqrt{(a-x)^2 + h_2^2} \cdot v_2} = 0$$

TRAŽIMO IZVOD
PO x I IZJEDNACUJEMO
SA NULOM

$$\frac{\sin \theta_1}{v_1} - \frac{\sin \theta_2}{v_2} = 0 \Rightarrow \frac{\sin \theta_1}{v_1} = \frac{\sin \theta_2}{v_2}$$

$$n = \frac{c}{v}$$

- BRZINA U VAKUUMU

- BRZINA U SREDINI

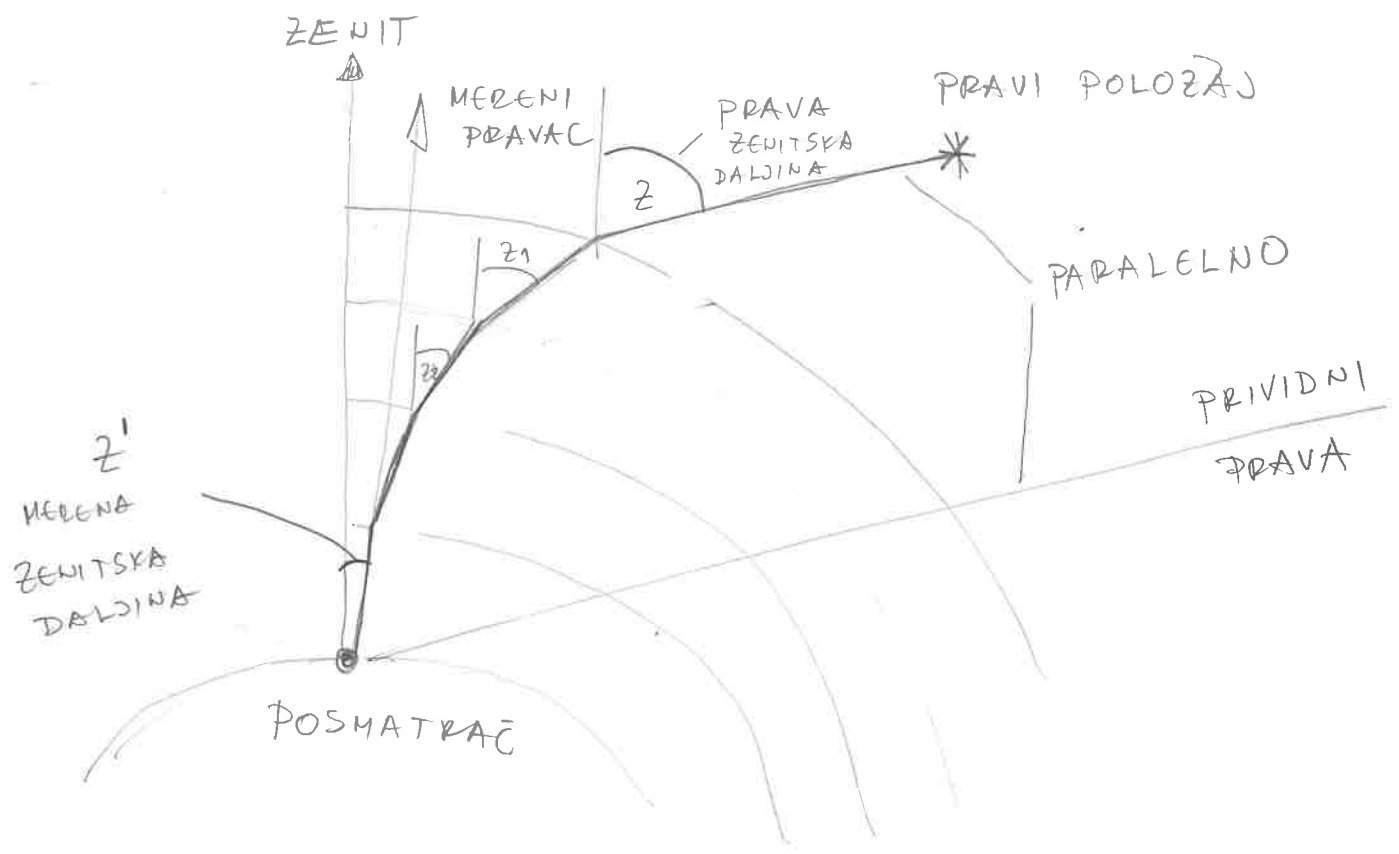
$$\frac{\sin \theta_1}{\frac{c}{n_1}} = \frac{\sin \theta_2}{\frac{c}{n_2}}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

REFRAKCIJA

- POJAVA KOJA DOVODI DO PRELAMANJA SVETLOSTI
PRI PRELAZU IZ JEDNE U DRUGU SREDINU
RAZLIČITIH KARAKTERISTIKA

- ASTRONOMSKA REFRAKCIJA PREDSTAVLJA
PRELAMANJE SVETLOSTI PRILIKOM PROLAŠKA
KROZ ZEMLJINU ATMOSFERU



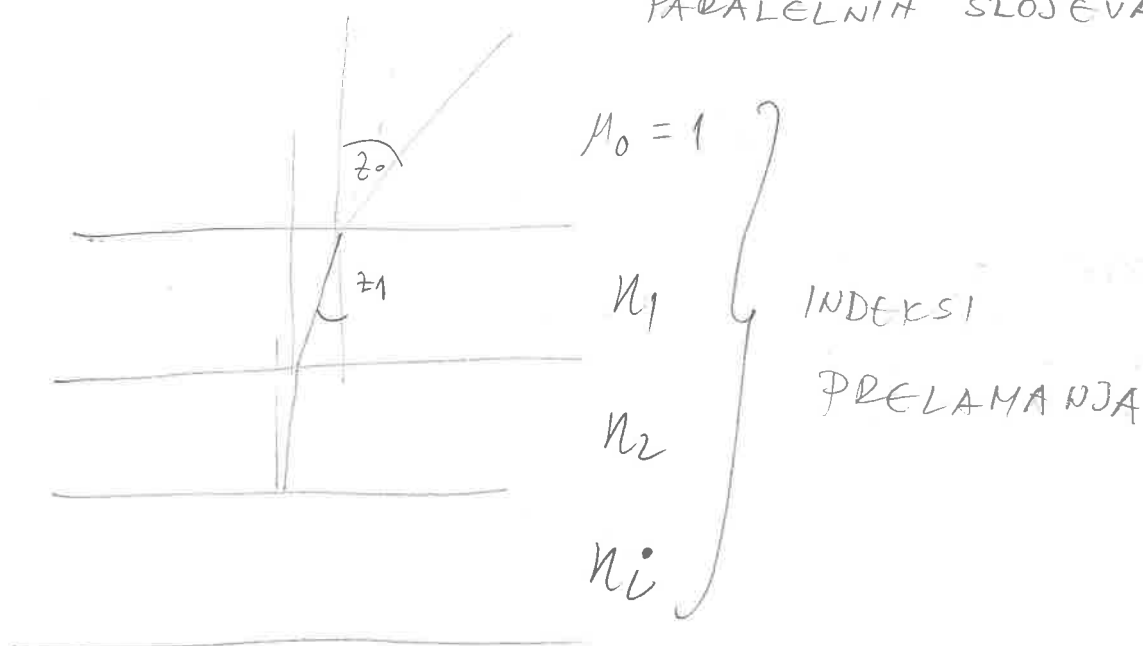
REFRAKCIJA DOVODI SAMO DO PROMENE
(SMANJENJA) ZENITSKE DALJINE, DOK NE UTIČE
NA AZIMUT.

$$\overset{\text{PRAVA}}{z} = \overset{\text{MERENA}}{z'} + \overset{\text{REFRAKCIJA}}{q}$$

$$\overset{\text{PRAVA}}{A} = \overset{\text{MERENA}}{A'}$$

- BUDUĆI DA SE REFRAKCIJA ODVIJA U

DONJIM SLOJEVIMA ATMOSFERE I BLIZU ZEMIT
MOŽEMO ATMOSFERU APROKSIMIRATI NIZOM RAVNIH
PARALELNIH SLOJEVA



$$\sin z_i n_i = \sin z_{i+1} n_{i+1}$$

FERMAOU
PRINCIP

Ako se prethodni izraz napiše za niz slojeva atmosfere, dobije se

$$\underbrace{n_0 \sin z}_{\text{izvan atmosfere}} = n_1 \sin z_1 = n_2 \sin z_2 = \dots = \underbrace{n' \sin z'}_{\text{posmatrač}}$$

pošto je $n_0 = 1$ (nema atmosfere)

$$\Rightarrow \sin z = n' \sin z'$$

Dakle, dovoljno je poznavati indeks prelamanja najnižeg sloja atmosfere n' .

$$\Rightarrow \sin(z' + \vartheta) = n' \sin z'$$

REFRAKCIJA

pošto je ϑ mali ugao, koristimo aproksimaciju

$$\sin \vartheta = \vartheta \quad (\vartheta \text{ je u radianima})$$

Ako želimo ϑ da izrazimo u lučnim sekundama

$$\sin \vartheta = \vartheta \cdot \frac{1''}{180 \cdot 3600} \quad \text{— JEDNA LUČNA SEKUNDA IZRAŽENA U RADIANIMA}$$

pošto je $1'' \approx \sin(1'')$ (ako se ugao izrazi u radianima)

$$\Rightarrow \sin \vartheta = \vartheta \sin(1'')$$

$$\sin(z' + \vartheta) = \sin z' \cos \vartheta + \cos z' \sin \vartheta$$

POŠTO JE ϑ MALI UGAO APROKSIMIRAMO $\cos \vartheta = 1$

$$\Rightarrow \sin z' + \cos z' \vartheta \sin(1'') = n' \sin z'$$

$$\Rightarrow \vartheta = \frac{n' - 1}{\sin(1'')} \operatorname{tg} z' = k \cdot \operatorname{tg} z'$$

ZA NORMALNE ATMOSFERSE USLOVE

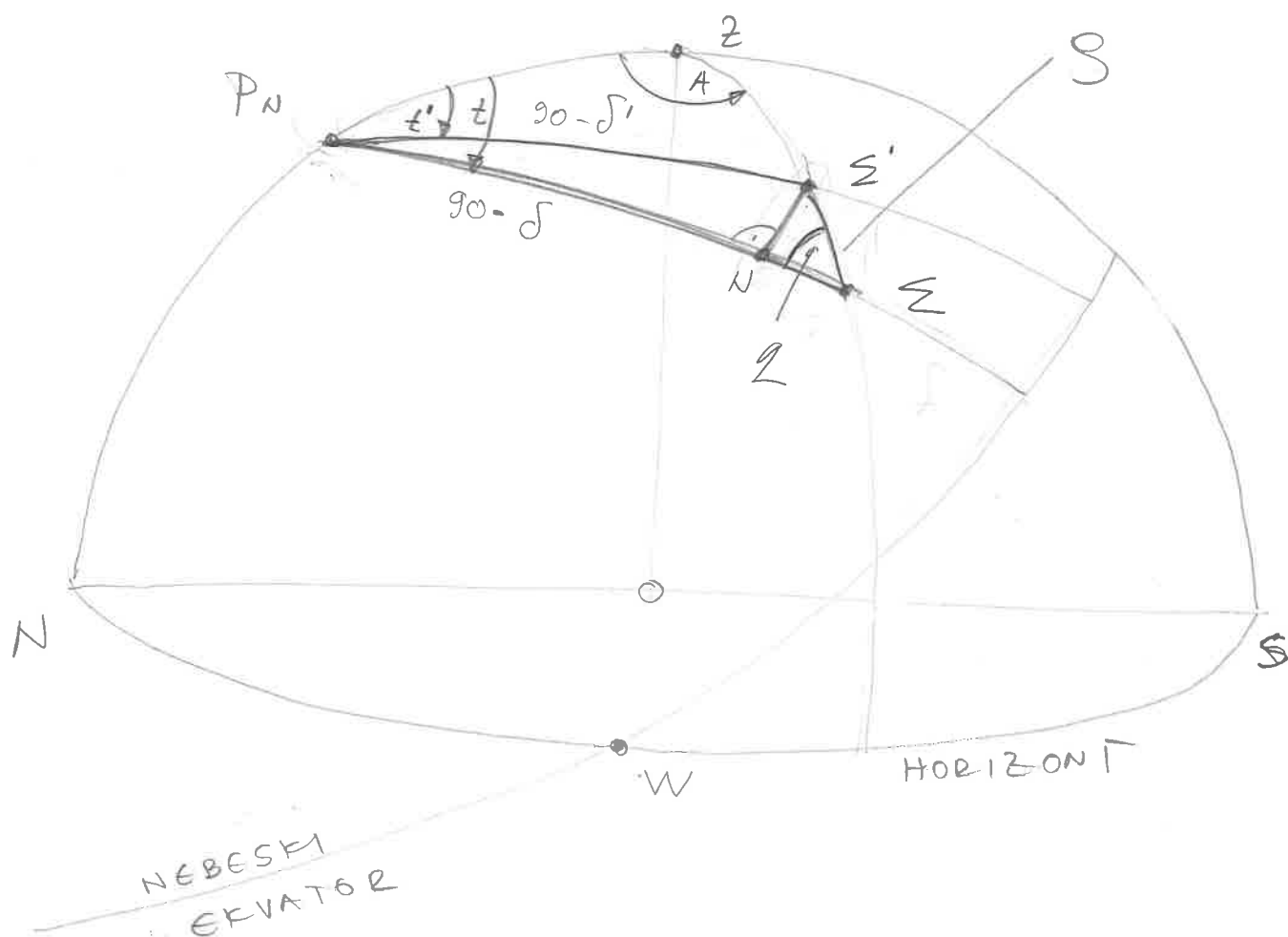
$$k = 60'', 154$$

KONSTANTA
ASTRONOMSKE
REFRAKCIJE

VREDNOST REFRAKCIJE DOBIJEN NA OVAJ NAČIN
NAZIVA SE SREDNJA (NORMALNA) REFRAKCIJA

ZA PRECIZNIJE RAČUNE UVODE SE POPRAVKE
POPUT TERMOMETARSKOG I BAROMETARSKOG FAKTORA
REFRAKCIJE.

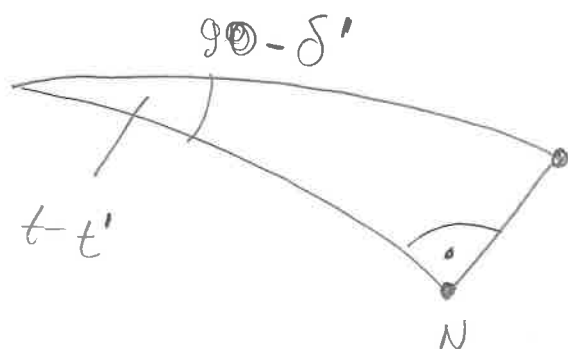
UTICAJ REFRAKCIJE NA EKVATORSKE KOORDINATE



$$N\Sigma' = S \sin z = k \operatorname{tg} z' \sin z$$

$$\delta - \delta' = N\Sigma = S \cos z = k \operatorname{tg} z' \cos z$$

SINUSNI OBRAZAC

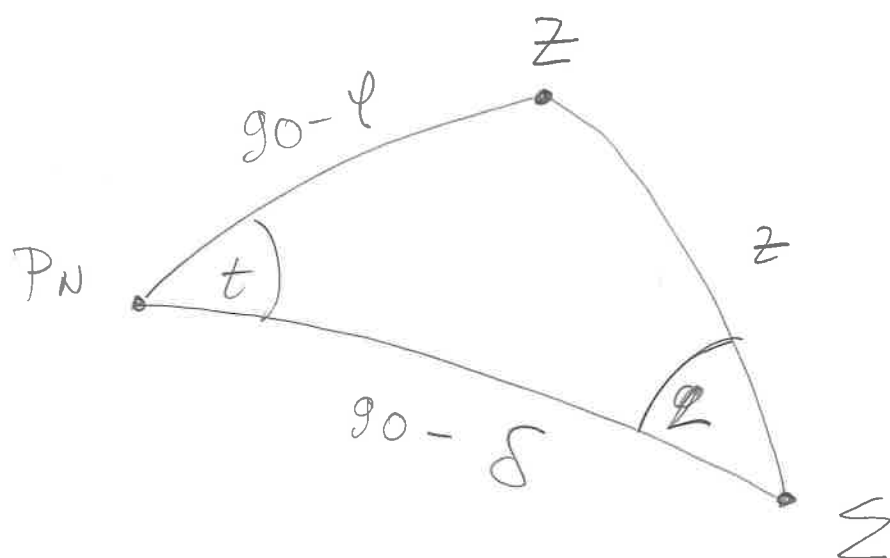


$$\frac{\sin N\Sigma'}{\sin(t-t')} = \sin(90-\delta')$$

$$N\Sigma' = (t-t') \cos \delta'$$

$$\Rightarrow \left[\delta - \delta' = k \frac{\sin z'}{\cos z'} \cos z \quad t - t' = k \frac{\sin z'}{\cos z'} \frac{\sin z}{\cos \delta'} \right]$$

U OBA IZRAZA FIGURISE UGAO 2



KOSINUSNI OBRAZAC ZA STRANICU z

$$\cos z = \cos(90-\varphi) \cos(90-\delta) + \sin(90-\delta) \sin(90-\varphi) \cos t$$

$$\cos z = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos t$$

SINUSNI OBRAZAC

$$\frac{\sin z}{\sin(90-\varphi)} = \frac{\sin t}{\sin z} \Rightarrow \sin z \sin z = \sin t \cos \varphi$$

SINUSNO - KOSINUSNI OBRAZAC

$$\sin z \cos z = \sin \varphi \cos \delta - \cos \varphi \sin \delta \cos t$$

ODACIMO U IZRAZE ZA $t-t'$ I $\delta-\delta'$ UZIMAJUCI
U OBZIR DA JE $z' \approx z$

$$t - t' = K \frac{\cos \varphi \sin t}{(\sin \delta \sin \varphi + \cos \delta \cos \varphi \cos t) \cos \delta}$$

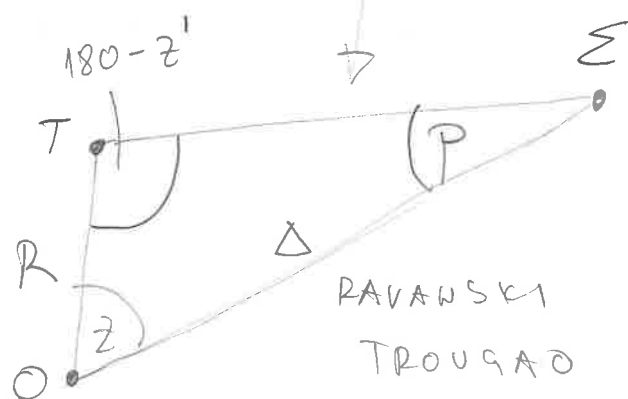
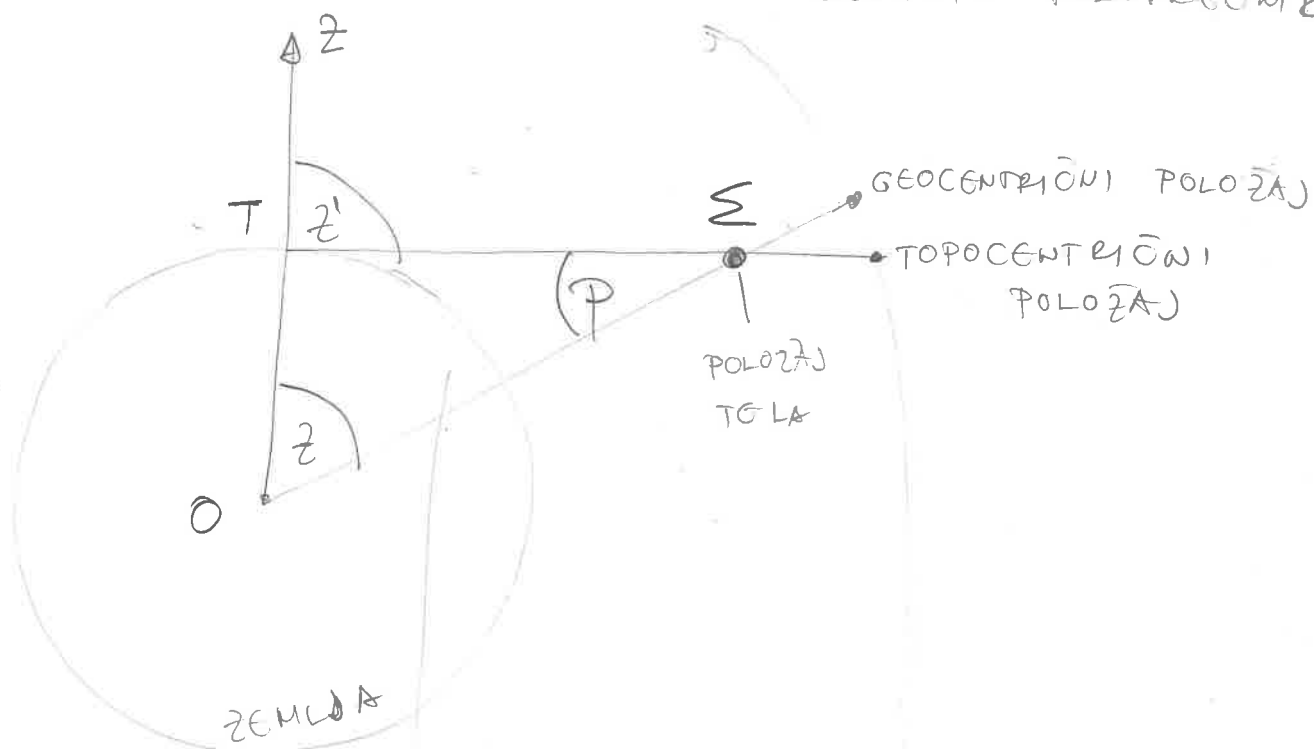
$$\delta - \delta' = K \frac{\sin \varphi \cos \delta - \cos \varphi \sin \delta \cos t}{\sin \delta \sin \varphi + \cos \delta \cos \varphi \cos t}$$

PARALAKSA

- PARALAKSA JE UGAO POD KOJIM SE IZ NEKE TAČKE VIDI NEKA DUŽ



DNEVNA PARALAKSA JE UGAO POD KOJIM SE SA NEKOG TELA VIDI ZEMLJIN POLUPREČNIK



NEBESKA SFERA

SLIKA 1

$$\frac{\sin p}{R} = \frac{\sin (180 - z')}{\Delta} \Rightarrow \frac{\sin p}{R} = \frac{\sin z'}{\Delta} \quad (1)$$

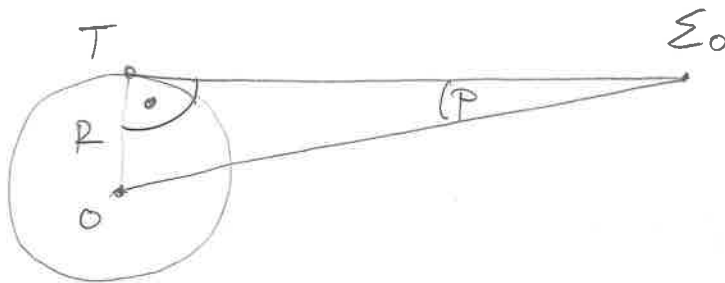
$\swarrow$
 POLUPREČNIK
 ZEMLJE

Δ —
 GEOCENTRIČNO
 RASTOJANJE

$$180 - z' + z + p = 180^\circ \Rightarrow z' = z + p \quad (2)$$

$$(1) \Rightarrow \sin p = \frac{R}{\Delta} \sin z'$$

— KADA JE $z' = 90^\circ$



SLIKA 2

$$p_0 \approx \sin p_0 = \frac{R}{\Delta}$$

Ako p_0 izrazimo u lučnim sekundama

onda je $\sin p_0 \approx p_0 \cdot \sin 1''$

$$p_0 \approx \frac{R}{\Delta \sin 1''} \Rightarrow \frac{R}{\Delta} = p_0 \sin 1''$$

PONOVO ANALIZIRAMO TROUGAO TOΣ

$$\sin P = \frac{R}{\Delta} \sin z'$$

$\swarrow$ $\searrow$
 $P \cdot \sin 1''$ $P_0 \sin 1''$

$$\Rightarrow \boxed{P = P_0 \sin z'}$$

UTICAJ DNEVNE PARALAKSE NA HORIZONTSKO KOORDINATE

$$A' = A$$

$$z' = z + P = z + P_0 \sin z'$$

DAKLE, DNEVNA PARALAKSA UVEK POVEĆAVA ZENITSKU DALJINU (SUPROTNO OD REFRAKCIJE).

- TREBA UZETI U OBZIR DA ZEMLJA NIJE SFERA TJ. DA SE POLUPREČNICI RAZLIKUJU U ZAVISNOSTI OD TOPOCENTRA

$$\sin P_0 = \frac{R}{\Delta} \sin z'$$

$$\sin P_e = \frac{R_e}{\Delta} \sin z'$$

~~~~~  
EKVATORSKA  
PARALAKSA

← EKVATORSKI POLUPREČNIK

# UTICAJ SPLOŠNOSTI ZEMLJE NA PROMENU

## PARALAKSE

$$\left. \begin{array}{l} \Delta P_{\text{MESEC}} \approx 6' 17'' \\ \Delta P_{\text{SUNCE}} \approx 0'',03 \end{array} \right\} \begin{array}{l} \text{RAZLIKE DNEVNIH PARALAKSI} \\ \text{KADA SE KAO REFERENTNE} \\ \text{DUŽINE UZIMAJU POLARNI} \\ \text{I EKVATORSKI POLUPREČNIK} \\ \text{ZEMLJE} \end{array}$$

- OBICNO SE DAJE UTABLIČENA EKVATORSKA DNEVNA PARALAKSA, PA SE PARALAKSA ZA ODGOVARAJUĆU ZENITSKU DALJINU DOBIJA IZ PRETHODNO IZVEDENOG IZRAZA

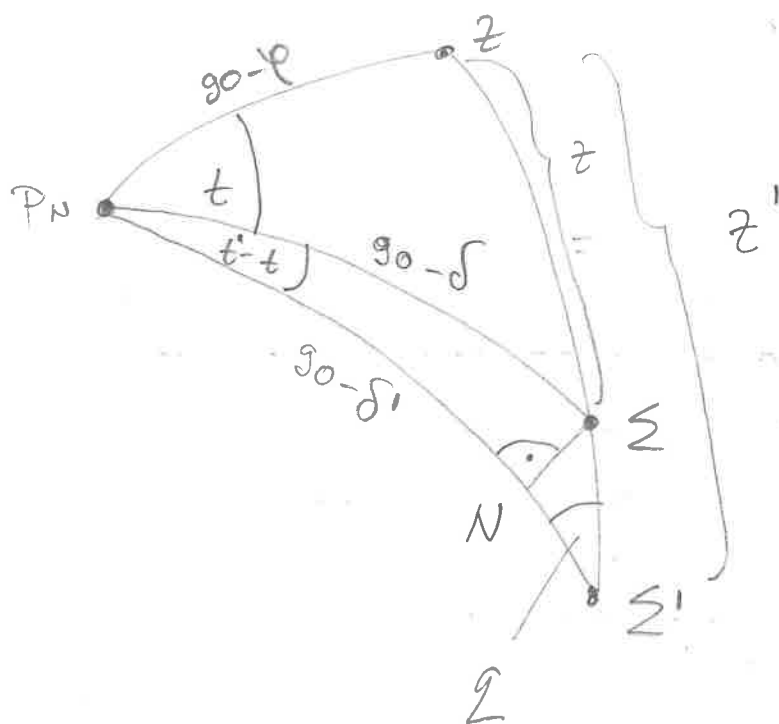
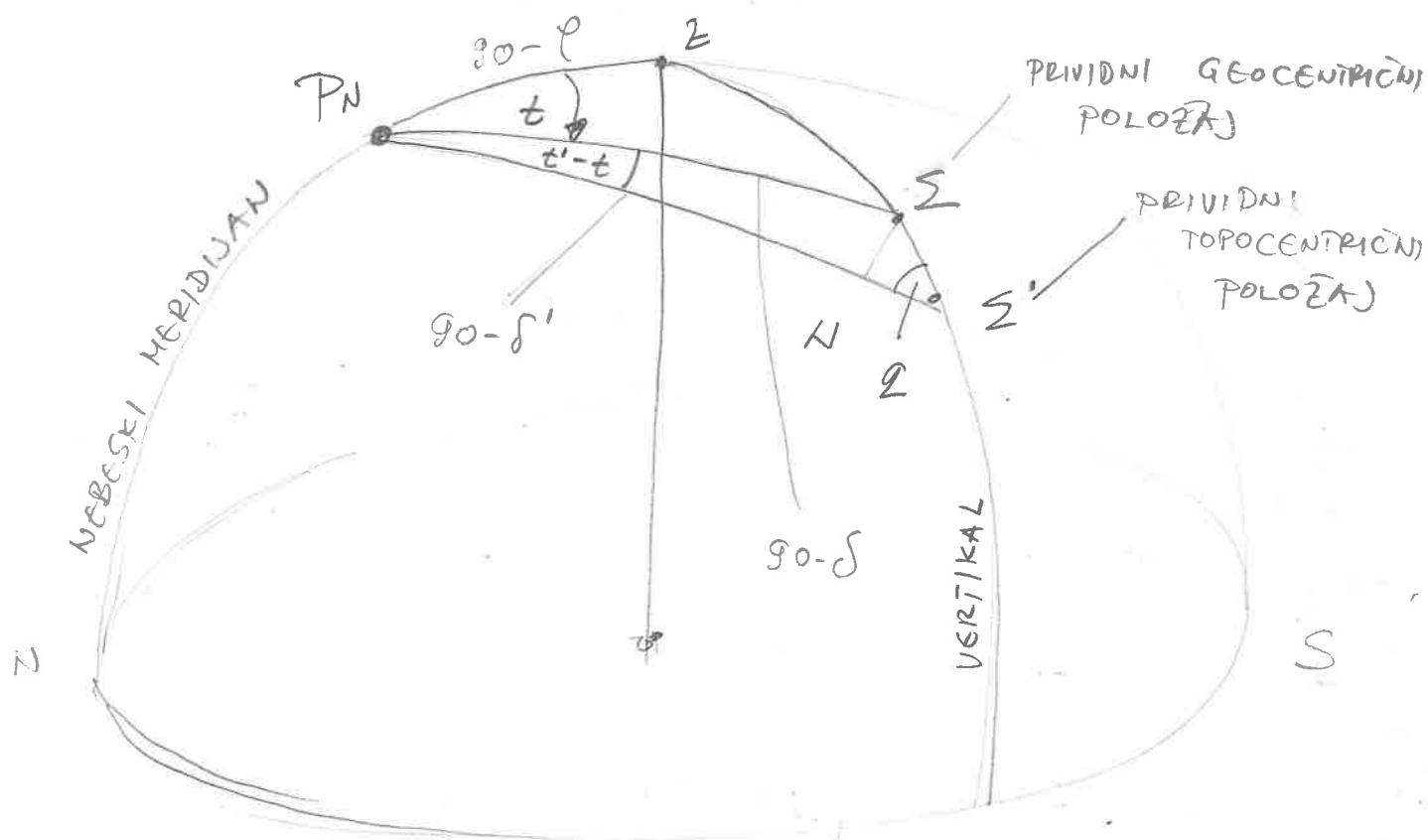
$$P = P_0 \sin z'$$

$$P_e = \frac{R_e}{\Delta \sin 1''} \quad - \quad \text{EKVATORSKA PARALAKSA}$$

- ZA PROIZVOLJNU R

$$P = \frac{R}{\Delta \sin 1''} = \frac{R}{R_e} \frac{R_e}{\Delta \sin 1''} = \frac{R}{R_e} P_e$$

# UTICAJ DNEVNE PARALAKSE NA EKVATORSKE KOORDINATE



SLIKA 3

$$\Delta N \Sigma \Sigma'$$

$$N \Sigma = \underbrace{\Sigma \Sigma'}_{P_0 \sin z'} \sin \varrho = P_0 \sin z' \sin \varrho$$

$$N \Sigma' = \Sigma \Sigma' \cos \varrho = P_0 \sin z' \cos \varrho$$

(1)

$$\Delta P N \Sigma N$$

SINUSNI OBRAZAC:

$$\frac{\sin N \Sigma}{\sin (t-t')} = \sin (90^\circ - \delta) = \cos \delta$$

$$\Rightarrow \sin N \Sigma = \sin (t-t') \cos \delta$$

POŠTO SU U PITANJU MALI UGLOVI

$$\Rightarrow N \Sigma = (t-t') \cos \delta \quad (2)$$

SA SLIKE SE VIDI

$$N \Sigma' = (90^\circ - \delta') - (90^\circ - \delta) = \delta - \delta' \quad (3)$$

IZJEDNACIMO (1) I (3)

$$\delta - \delta' = P_0 \sin z' \cos \varrho \quad (4)$$

IZJEDNACIMO (1) I (2)

$$(t'-t) \cdot \cos \delta = P_0 \sin z' \sin \varrho \quad (5)$$

U IZRAZIMA (4) I (5) FIGURISE UGAO  $\varphi$ .  
 OVAJ UGAO ĆEMO ODREDITI IZ SFERNOG  
 TROUGLA  $\triangle P_N Z \Sigma'$  (SLIKA 3)

SINUSNI OBRAZAC

$$\frac{\sin z'}{\sin t'} = \frac{\sin(90-\varphi)}{\sin \varphi} = \frac{\cos \varphi}{\sin \varphi}$$

$$\sin z' \sin \varphi = \cos \varphi \sin t' \quad (6)$$

SINUSNO - KOSINUSNI OBRAZAC

$$\sin z' \cos \varphi = \cos(90-\varphi) \sin(90-\delta') -$$

$$\sin(90-\varphi) \cos(90-\delta') \cos t'$$

$$\sin z' \cos \varphi = \sin \varphi \cos \delta' - \cos \varphi \sin \delta' \cos t' \quad (7)$$

UZIMAJUĆI U OBZIR DA JE

$$\cos \delta' \approx \cos \delta$$

$$\sin t' \approx \sin t$$

UVRŠTAVANJEM (7) U (4) DOBJAMO

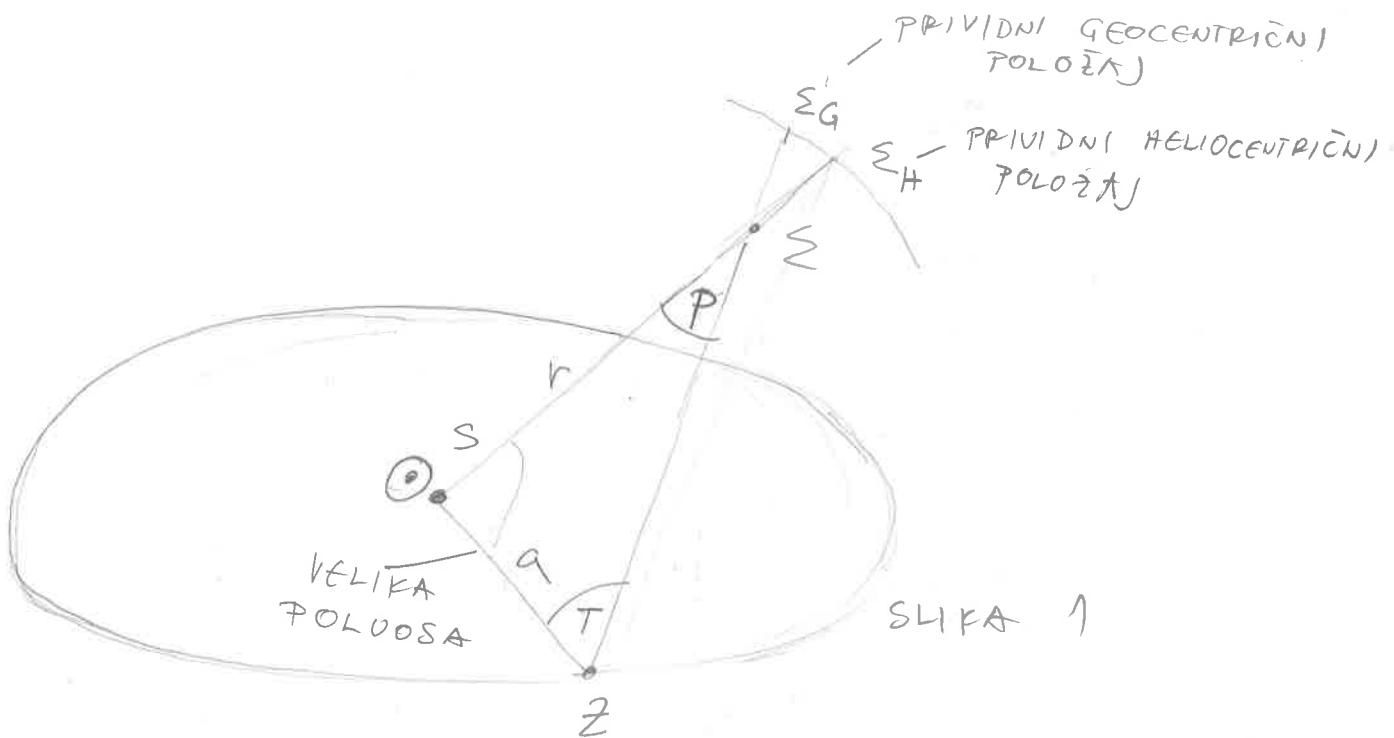
$$\delta' - \delta = -p_0 (\sin \varphi \cos \delta - \cos \varphi \sin \delta \cos t)$$

UVRŠTAVANJEM (6) U (5) DOBJAMO

$$t' - t = p_0 (\sec \delta \cos \varphi \sin t)$$



# GODISNJA PARALAKSA



$$\triangle SZ\Sigma \Rightarrow \frac{\sin P}{a} = \frac{\sin T}{r}$$

$$\boxed{\sin P = \frac{a}{r} \sin T} \quad (1)$$

AKO  $P$  IZRAŽIMO U LUČNIM SEKUNDAMA

$$\Rightarrow \sin P (\text{rad}) \approx P (\text{rad}) = P (") \cdot \frac{1}{206264,8}$$

JEDNA LUČNA SEKUNDA  
IZRAŽENA U RADIJANIMA

POSTO ZA MALI UGAO  $\alpha \Rightarrow \sin \alpha (\text{rad}) \approx \alpha (\text{rad})$

$$\Rightarrow \frac{1}{206264,8} \approx \sin 1''$$

$$\Rightarrow P (\text{rad}) \approx P (") \cdot \sin 1''$$

KADA SE OVO UVRSTI U (1)  $\Rightarrow$

KAO I

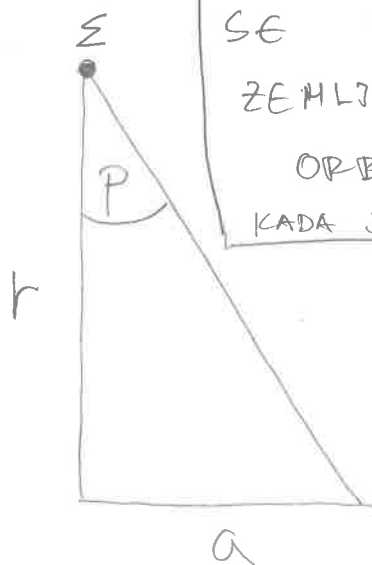
$$\sin P \approx P$$

$$\Rightarrow \boxed{P = \frac{a}{r \sin 1''} \sin T}$$

$\pi$  — GODIŠNJA PARALAKSA

$$P = \pi \sin T$$

GODIŠNJA PARALAKSA  
NEBESKOG TELA . PREDSTAVLJA  
NAJVEĆI UGAO POD KOJIM  
SE VIDI VELIKA POLUOSA  
ZEMLJINE HELIOCENTRIČNE  
ORBITE !!!  
KADA JE  $T = 90^\circ$



$$\frac{a}{r} = \tan P \approx P$$

KADA JE  $P = 1''$  TADA JE

$$r = 1 \text{ pc}$$

DAKLE, RASTOJANJE OD JEDNOG PARSEKA  
ODGOVARA RASTOJANJU SA KOJEG SE JEDNA  
ASTRONOMSKA JEDINICA VIDI POD UGLOM OD  
JEDNE LUČNE SEKUNDE

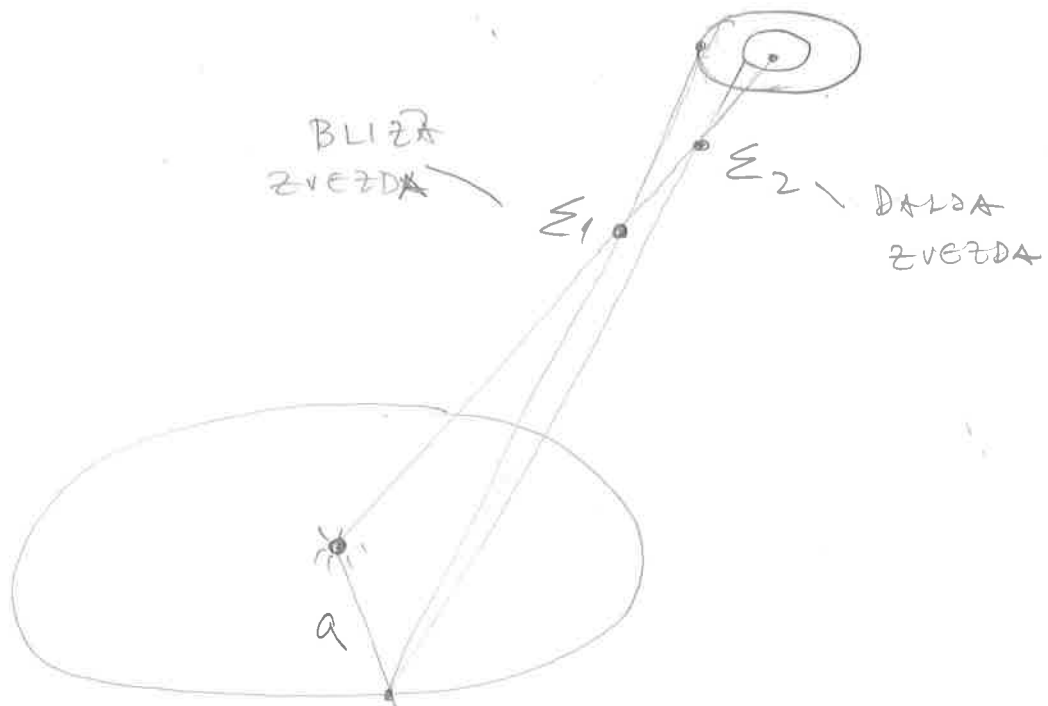
$$\Rightarrow 1 \text{ pc} = \frac{a}{\sin 1''} = d$$

SADA IZ  $p = \frac{a}{r \sin 1''} \sin T$  SLEDI

$$\pi'' = \frac{a}{r \sin 1''} = \frac{d}{r} = \frac{1}{r/d}$$

$$\pi'' = \frac{1}{r/d} \Rightarrow r(\text{pc}) = \frac{1}{\pi''}$$

RASTOJANJE IZRAŽENO U PARSECIMA  
 PREDSTAVLJA RECIPROČNU VREDNOST  
 GODIŠNJE PARALAKSE IZRAŽENE U LUČNIM  
 SEKUNDAMA !



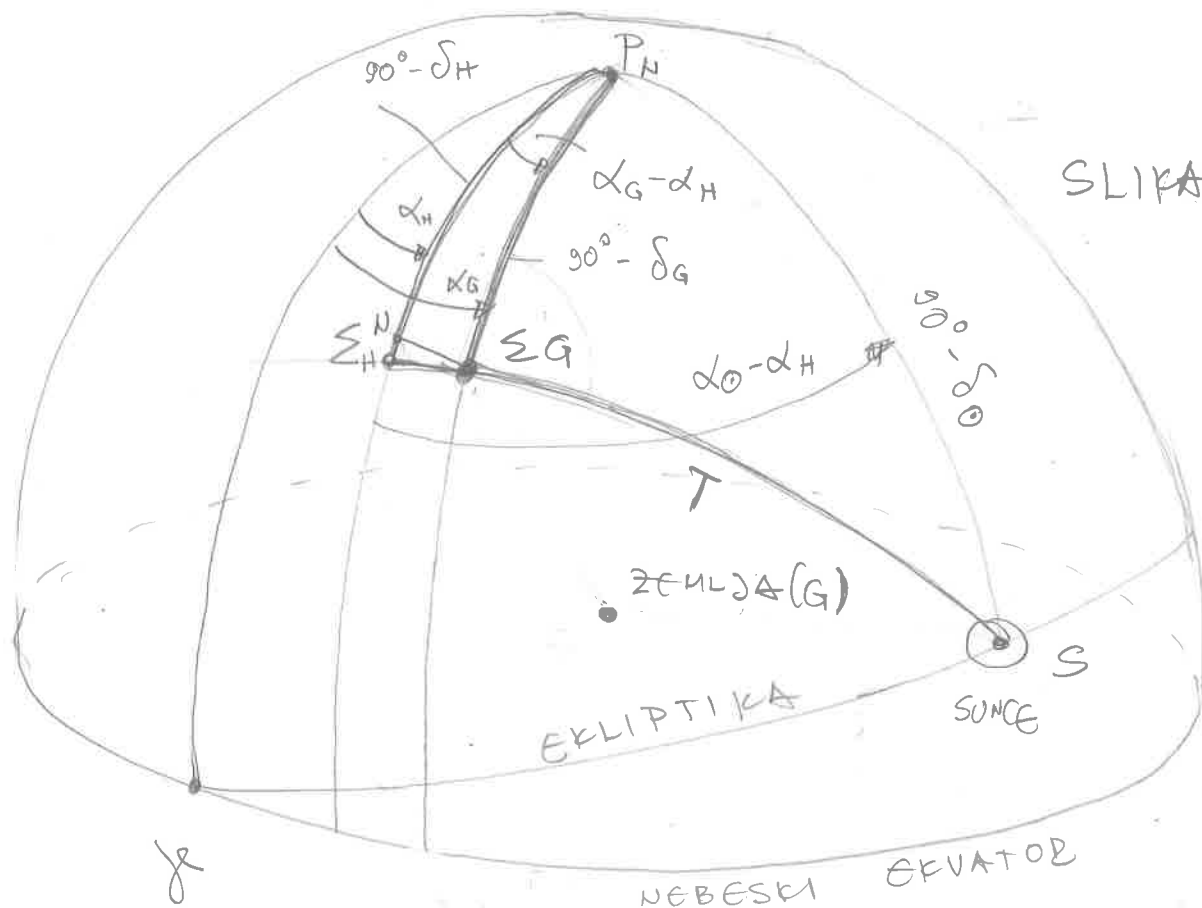
TOKOM GODIŠNJEG KRETANJA ZEMLJE OKO SUNCA ZVEZDE NA NEBESKOS SFERI OPISUJU TAKOZVANE PARALAKTIČKE ELIPSE ČIJE VELIKE POLUOSE SU JEDNAKE GODIŠNJOJ PARALAKSI ZVEZDE.

PARALAKTIČKE ELIPSE PREDSTAVLJAJU CENTRALNE PROJEKCIJE ZEMLJINE ORBITE NA NEBESKU SFERU.

UTICAJ GODIŠNJE PARALAKSE NA  
EKVATORSKE KOORDINATE

$\sum_H$  - HELIOCENTRIC POSITION

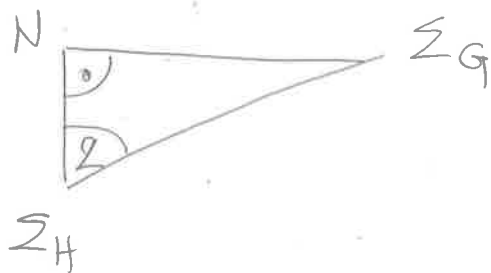
$\Sigma_G$  - GEOCENTRICNI POLOŽAJ



SLIPA 2.

PROVOGLI: TROUGAO  $\Delta \Sigma_H \Sigma_G N$

VEOMA JE MALI PA SE MOŽE SHATATI RAVANSKIM



$$\Sigma_G N = \Sigma_H \Sigma_G \sin \alpha$$

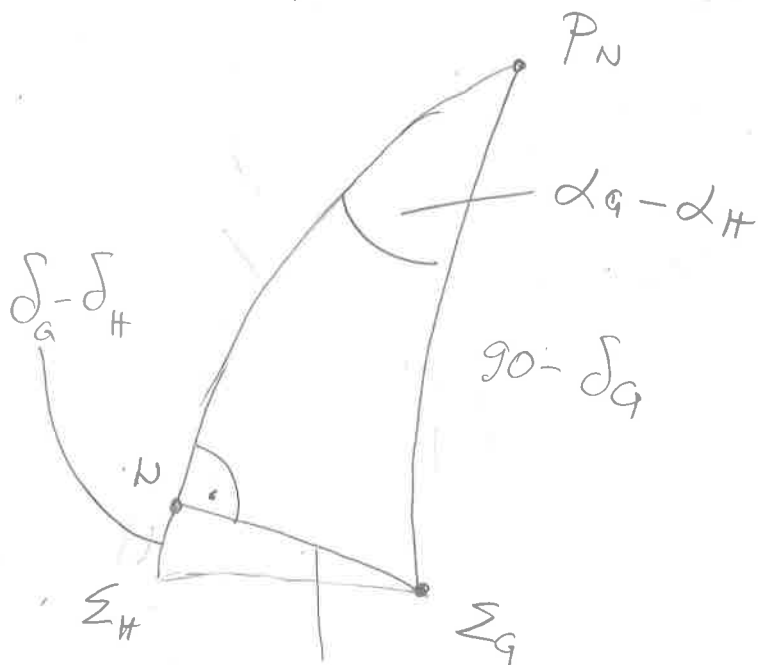
$$\sum_H N = \sum_H \sum_G \cos \theta$$

$$\Sigma_H \Sigma_g = \pi \sin(T)$$

1  
SLIKA 1

$$\left. \begin{aligned} \Sigma_G N &= \pi \sin T \sin \varrho \\ \Sigma_H N &= \pi \sin T \cos \varrho \end{aligned} \right\} (1)$$

$\Delta P_N \Sigma_G N$  — SFERNI PRAVOUTGLI



$$\Sigma_H \Sigma_G \sin \varrho = \pi \sin T \sin \varrho$$

SINUSNI OBRAZAC

$$\frac{\sin(N \Sigma_G)}{\sin(\alpha_G - \alpha_H)} = \sin(90^\circ - \delta_G) = \cos \delta_G$$

$$\sin(N \Sigma_G) = \sin(\alpha_G - \alpha_H) \cos \delta_G$$

MALE UGAO  $\Rightarrow \sin(N \Sigma_G) \approx N \Sigma_G$

$$N \Sigma_G = \sin(\alpha_G - \alpha_H) \cos \delta_G$$

$\alpha_G - \alpha_H$  JE MALE UGAO PA JE  $\sin(\alpha_G - \alpha_H) \approx \alpha_G - \alpha_H$

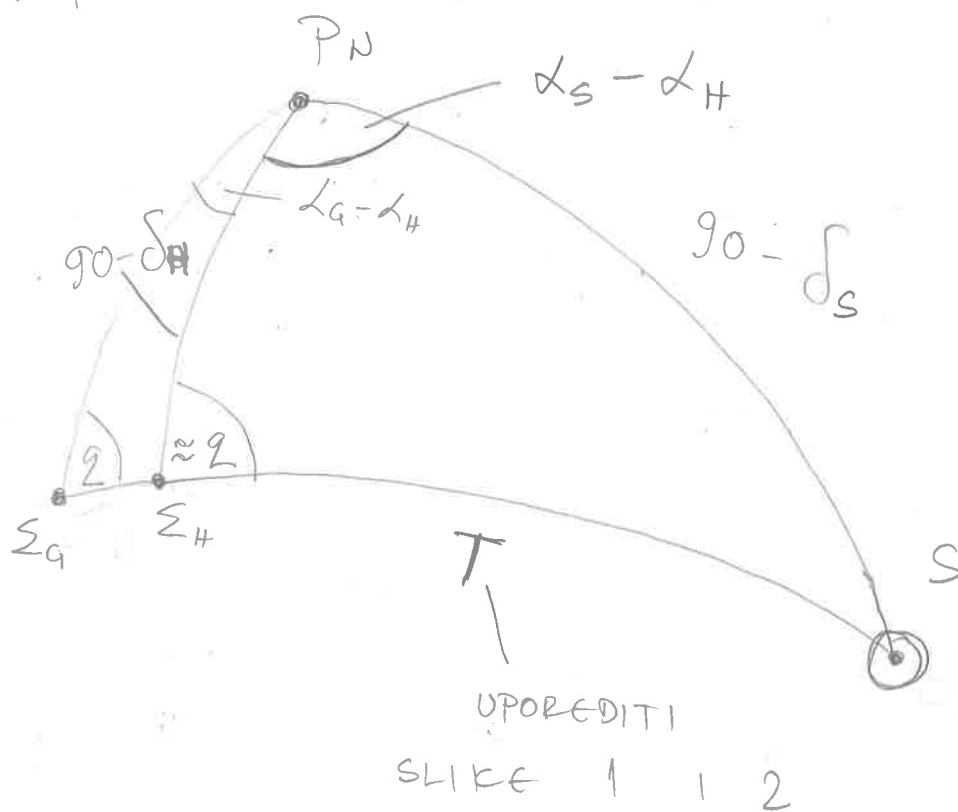
$$\Rightarrow \left. \begin{aligned} N \Sigma_G &= (\alpha_G - \alpha_H) \cos \delta_G \\ N \Sigma_H &= \delta_G - \delta_H \end{aligned} \right\} (2)$$

SADA SE (1) UVRSTI U (2)

$$\Rightarrow \left. \begin{aligned} (\alpha_G - \alpha_H) \cos \delta_G &= \pi \sin T \sin \varphi \\ \delta_G - \delta_H &= \pi \sin T \cos \varphi \end{aligned} \right\} (3)$$

POTREBNO JE DA NAĐEMO  $\sin T \sin \varphi$  I  $\sin T \cos \varphi$

$\triangle P_N \Sigma_G S$  (SLIKA 2)



SINUSNI OBRAZAC

$$\frac{\sin T}{\sin(\alpha_{\odot} - \alpha_H)} = \frac{\sin(90 - \delta_{\odot})}{\sin 2}$$

$$\sin T \sin 2 = \sin(\alpha_{\odot} - \alpha_H) \cos \delta_{\odot} \quad (4)$$

# SINUSNO - KOSINUSNI OBRAZAC

$$\sin T \cos \varphi = \cos(90^\circ - \delta_\odot) \sin(90^\circ - \delta_H) -$$

$$\sin(90^\circ - \delta_\odot) \cos(90^\circ - \delta_H) \cdot \cos(\alpha_\odot - \alpha_H)$$

$$\sin T \cos \varphi = \sin \delta_\odot \cos \delta_H - \cos \delta_\odot \sin \delta_H \cos(\alpha_\odot - \alpha_H) \quad (5)$$

$$(4), (5) \longrightarrow (3)$$

RAZLIKE GEOCENTRIČNIH I HELIOCENTRIČNIH KOORDINATA

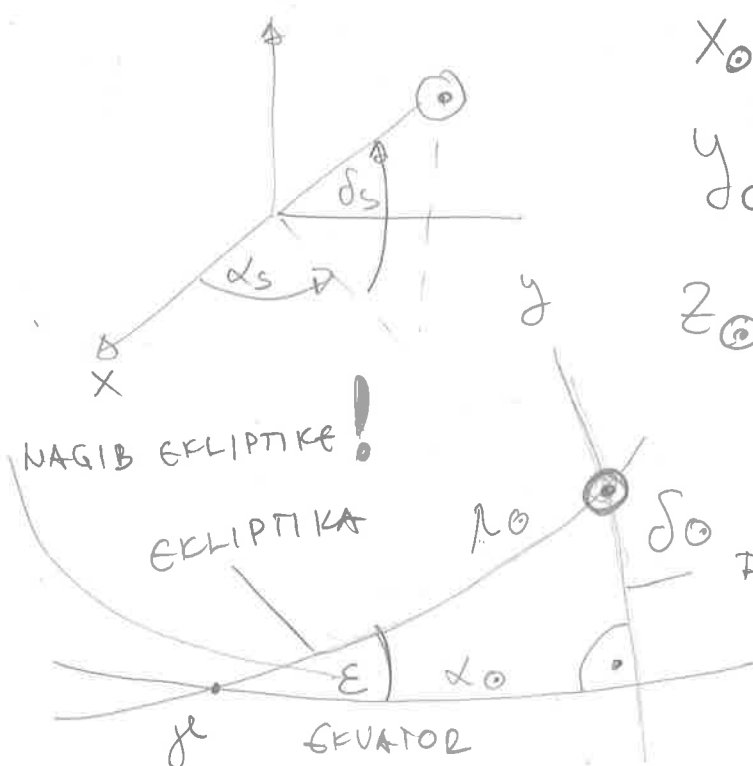
$$(6) \quad \begin{aligned} \alpha_G - \alpha_H &= \pi \sin(\alpha_\odot - \alpha_H) \cos \delta_\odot \sec \delta_G \\ \delta_G - \delta_H &= \pi \left[ \sin \delta_\odot \cos \delta_H - \cos \delta_\odot \sin \delta_H \cos(\alpha_\odot - \alpha_H) \right] \end{aligned}$$

OVA RAZLIKA KOORDINATA MOŽE DA SE IZRAZI  
I U FUNKCIJI PRAVOUGLIH EKLIPTIČKIH KOORDINATA  
SUNCA.

$$X_\odot = \cos \delta_\odot \cos \alpha_\odot$$

$$Y_\odot = \cos \delta_\odot \sin \alpha_\odot$$

$$Z_\odot = \sin \delta_\odot$$



DEKLINACIJSKI KRUG



SINUSNI OBRATAC

$$\frac{\sin \delta_0}{\sin \varepsilon} = \sin \lambda_0 \Rightarrow \sin \delta_0 = \sin \varepsilon \sin \lambda_0$$

SINUSNO - KOSINUSNI OBRATAC

$$\sin \lambda_0 \cos \varepsilon = \sin \alpha_0 \cos \delta_0 - \cos \alpha_0 \sin \delta_0 \cos 90^\circ$$

$$\sin \lambda_0 = \frac{\sin \alpha_0 \cos \delta_0}{\cos \varepsilon}$$

UVRSTI SG  
U SINUSNI  
OBRATAC

$$\sin \delta_0 = \operatorname{tg} \varepsilon \underbrace{\sin \alpha_0 \cos \delta_0}_{y_0} \quad (7)$$

$$\sin \delta_0 = \operatorname{tg} \varepsilon y_0$$

$$\sin(\alpha_0 - \alpha_H) = \sin \alpha_0 \cos \alpha_H - \cos \alpha_0 \sin \alpha_H$$

Iz JEDNAČINE (6)

$$\alpha_G - \alpha_H = \pi \sin(\alpha_0 - \alpha_H) \cos \delta_0 \sec \delta_G$$

$$= \pi \left( \underbrace{\sin \alpha_0 \cos \delta_0 \cos \alpha_H}_{y_0} - \underbrace{\cos \alpha_0 \cos \delta_0 \sin \alpha_H}_{x_0} \right) \sec \delta_G$$

$$\alpha_G - \alpha_H = \pi \left[ y_0 \sec \delta_H \cos \alpha_H - x_0 \sec \delta_H \sin \alpha_H \right]$$

$$C = \frac{1}{15} \sec \delta_H \cos \alpha_H$$

$$\delta = \frac{1}{15} \sec \delta_H \sin \alpha_H$$

BESELOVE  
KONSTANTE

$\frac{1}{15}$  JE ZBOG TOGA ŠTO U JEDNOJ VREMENSKOJ  
SEKUNDI IMA 15 LUCNIH SEKUNDI

$$\alpha_G - \alpha_H = \pi (y_0 C - x_0 d)$$

$$\delta_G - \delta_H = \pi (y_0 C' - x_0 d')$$

(8)

IZ JEDNAČINE (6)

$$\cos(\alpha_0 - \alpha_H) = \cos \alpha_0 \cos \alpha_H + \sin \alpha_0 \sin \alpha_H$$

$$\delta_G - \delta_H = \pi (\sin \delta_0 \cos \delta_H - \cos \delta_0 \sin \delta_H \cos(\alpha_0 - \alpha_H))$$

$$= \pi \left( \underbrace{\sin \delta_0}_{\text{tg } \epsilon y_0} \cos \delta_H - \cos \delta_0 \sin \delta_H \underbrace{\cos \alpha_0}_{x_0} \cos \alpha_H - \right.$$

$$\left. \cos \delta_0 \sin \delta_H \underbrace{\sin \alpha_0}_{y_0} \sin \alpha_H \right)$$

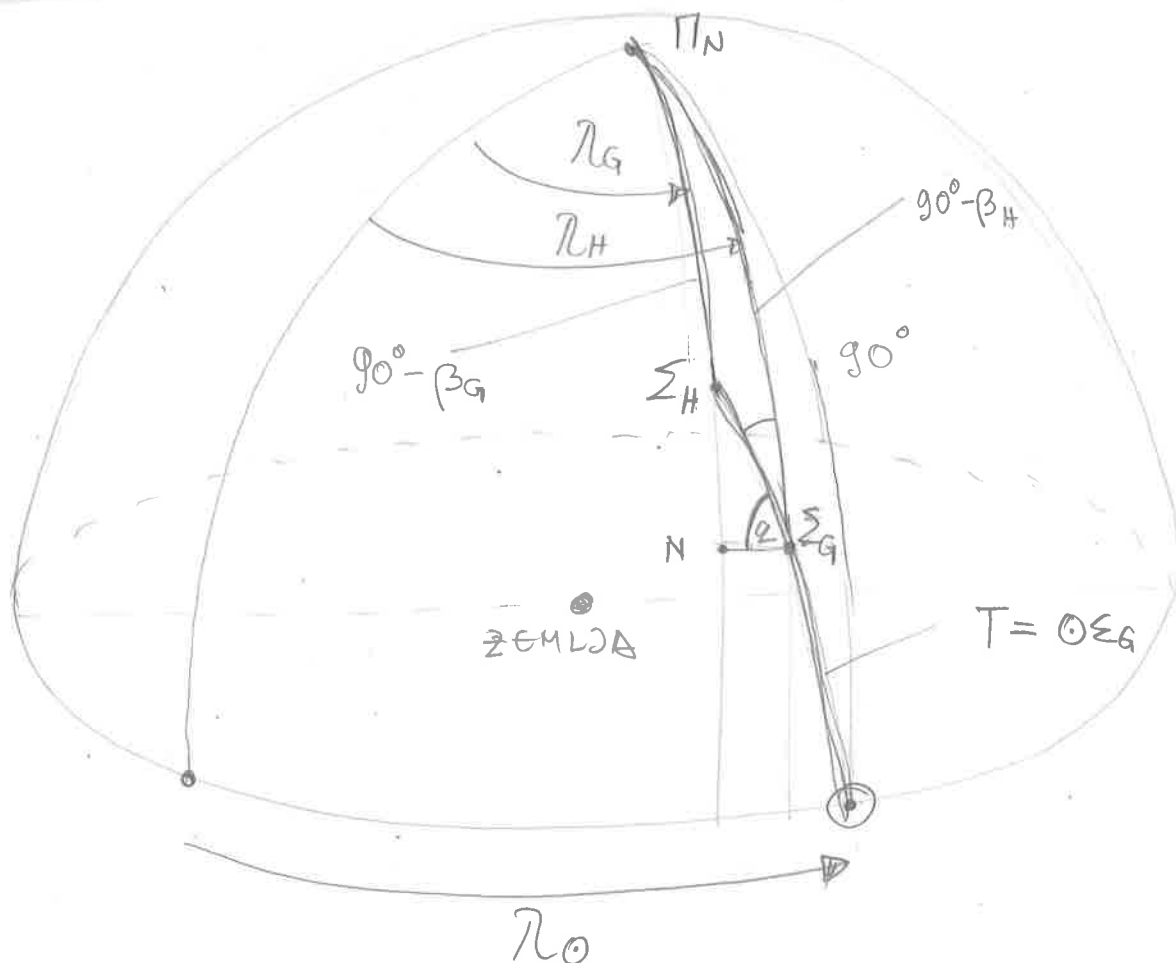
$$= \pi \left[ y_0 \text{tg } \epsilon \cos \delta_H - x_0 \sin \delta_H \cos \alpha_H - y_0 \sin \delta_H \sin \alpha_H \right]$$

$$= \pi \left[ y_0 \underbrace{(\text{tg } \epsilon \cos \delta_H - \sin \delta_H \sin \alpha_H)}_{C'} - x_0 \underbrace{\sin \delta_H \cos \alpha_H}_{d'} \right]$$

$$C' = \text{tg } \epsilon \cos \delta_H - \sin \delta_H \sin \alpha_H$$

$$d' = \sin \delta_H \cos \alpha_H$$

GODIŠNJE PARALAKSE NA  
EKLIPTIČKE KOORDINATE



$\Delta \Sigma_G \Sigma_H N$  - MALI SFERNI TROUGAO KOJI SE MOŽE SMATRATI RAVANSKIM

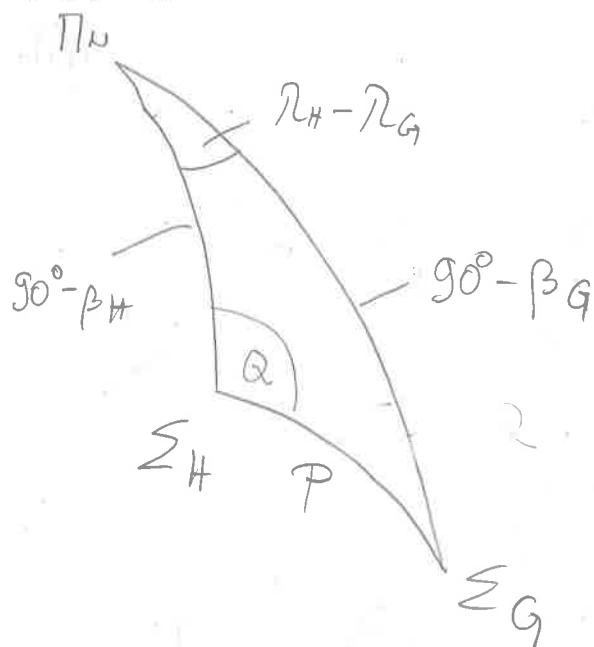
$$\Sigma_G \Sigma_H = P = \hat{\Pi} \sin T$$

$$N \Sigma_{\#} = \Sigma_G \Sigma_H \cos \varrho = \pi \sin T \cos \varrho$$

$$N \varepsilon_G = \varepsilon_G \varepsilon_H \sin \varphi = \pi \sin T \sin \varphi$$

SFERNI TROUGAO

$\Pi_N \Sigma_H \Sigma_G$



$$\frac{\sin(\lambda_H - \lambda_G)}{\sin P} = \frac{\sin Q}{\sin(90^\circ - \beta_H)}$$

$$\sin(\lambda_H - \lambda_G) \sin(90^\circ - \beta_G) = \sin P \sin Q$$

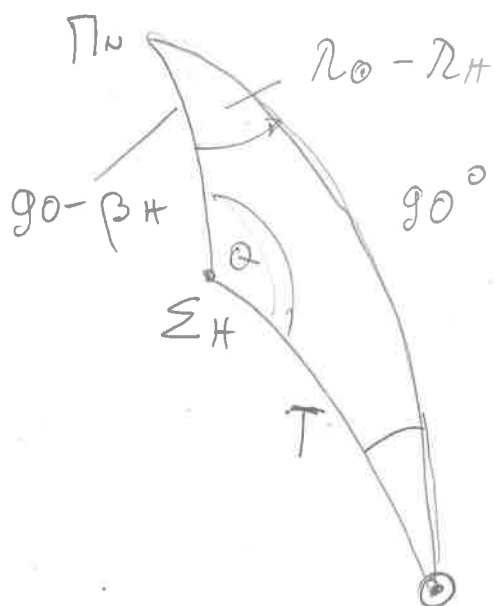
OVASTO  $\beta_G$  MOŽE DA SE PIŠE  $\beta_H$  ZBOG  
MALE RAZLIKE

$$\underbrace{\sin(\lambda_H - \lambda_G)}_{\approx \lambda_H - \lambda_G} \cos \beta_H = \sin P \sin Q$$

$$(\lambda_H - \lambda_G) \cos \beta_H = \sin P \sin Q = \Pi \sin T \sin Q \quad (9)$$

SFERNI TROUGAO

$\Delta \Pi_N \Sigma_H \odot$



SINUSNI OBRAZAC

$$\frac{\sin T}{\sin (\lambda_0 - \lambda_H)} = \frac{\sin 90^\circ}{\sin Q}$$

$\sin T \sin Q = \sin (\lambda_0 - \lambda_H) \rightarrow$  UVRSIT SE U (9),  
 RAZLIKA GEOCENTRICNE I HELIOCENTRICNE ECLIPTICKE LONGITUDE

$$\boxed{\lambda_H - \lambda_G = \pi \sin (\lambda_0 - \lambda_H) \sec \beta} \quad (10)$$

SINUSNO-KOSINUSNI OBRAZAC  $\Delta \Sigma_H \Sigma_G \Pi_N$

$$\sin p \cos Q = \cos (90^\circ - \beta_G) \sin (90^\circ - \beta_H) -$$

$$- \sin (90^\circ - \beta_G) \cos (90^\circ - \beta_H) \cos (\lambda_H - \lambda_G)$$

$$= \sin \beta_G \cos \beta_H - \cos \beta_G \sin \beta_H \cos (\lambda_H - \lambda_G)$$

POSTO JE  $\lambda_H - \lambda_G$  VEOMA MALI UGAO, PIŠEMO  
 DA JE  $\cos(\lambda_H - \lambda_G) = 1$

$\Rightarrow$

$$\sin p \cos Q = \sin \beta_G \cos \beta_H - \cos \beta_G \sin \beta_H = \sin(\beta_G - \beta_H)$$

POSTO JE  $\sin(\beta_G - \beta_H) \approx \beta_G - \beta_H$  PIŠEMO  
 $\sin p \approx p$

$$\beta_G - \beta_H = \sin p \cos Q = p \cos Q = \pi \frac{\sin T \cos Q}{\text{TRAŽIMO OVO}}$$

$$\Delta \Pi \Sigma_H \odot$$

SINUSNO-KOSINUSNI OBRAZAC

$$\begin{aligned} \sin T \cos Q &= \sin(90^\circ - \beta_H) \cos 90^\circ - \sin 90^\circ \cos(90^\circ - \beta_H) \cos(\lambda_0 - \lambda_H) \\ &= \sin \beta_H \cos(\lambda_0 - \lambda_H) \end{aligned}$$

PAŽLIKA GEOCENTRICNE I HELIOCENTRICNE ECLIPTIČKE LATITUDE

$\Rightarrow$

$$\boxed{\beta_G - \beta_H = \pi \sin \beta_H \cos(\lambda_0 - \lambda_H)} \quad (11)$$

ODREĐIVANJE DALJINE DO ZVJEZDE  
POMOĆU GODIŠNJE PARALAKSE

KORISTIMO JEDNAČINE (8)

$$\alpha_G - \alpha_H = \pi (y_0 c - x_0 d)$$

$c, d$  - BESELOVE KONSTANTE

MERIMO  $\alpha_G$  U RAZLIČITIM TRENUCIMA  
PREMA (8)

$$\alpha_{G_1} - \alpha_H = \pi \overbrace{(y_{01} c - x_{01} d)}^{A_1}$$

$$\alpha_{G_2} - \alpha_H = \pi \overbrace{(y_{02} c - x_{02} d)}^{A_2}$$

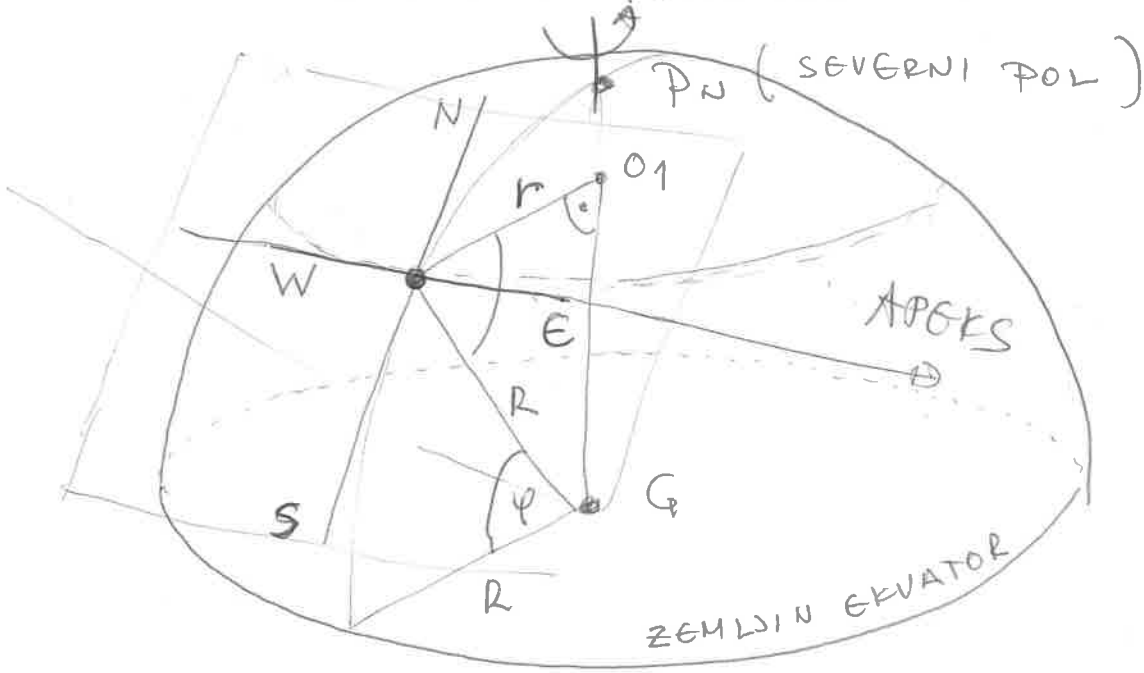
ODUŽMU SE PRETHODNE DVE JEDNAČINE

$$\Rightarrow \alpha_{G_2} - \alpha_{G_1} = \pi (A_2 - A_1)$$

$$\Rightarrow \pi = \frac{\alpha_{G_2} - \alpha_{G_1}}{A_2 - A_1}$$

# DNEVNA ABERACIJA

POSLEDICA ZEMLJINE POTACISE OKO SVOJE OSE



$$r = R \cos \varphi$$

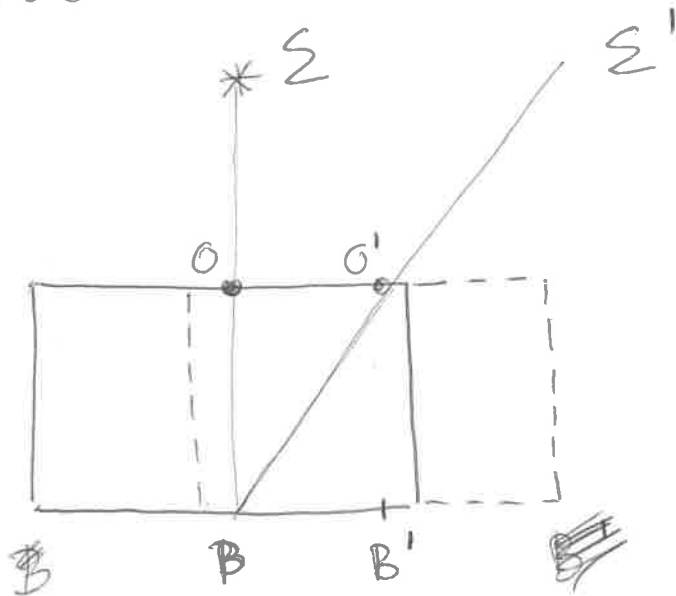
$$V_{\varphi} = \frac{2\pi r}{P} = \frac{2\pi R \cos \varphi}{P} = V_e \cos \varphi$$

P - PERIOD ROTACIDE ZEMLE

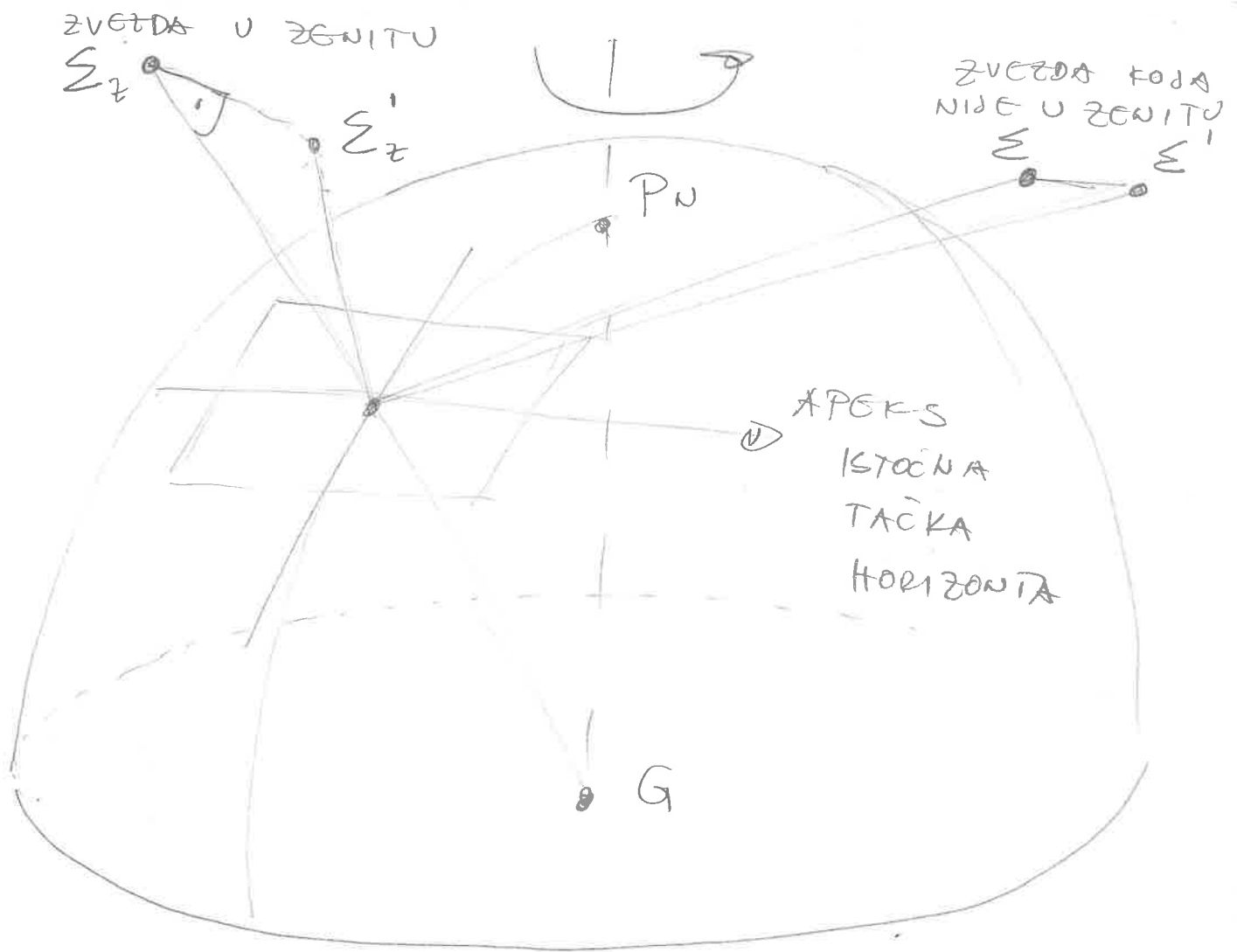
$V_e$  - TANGENCIJALNA BRZINA NA EKVATORU



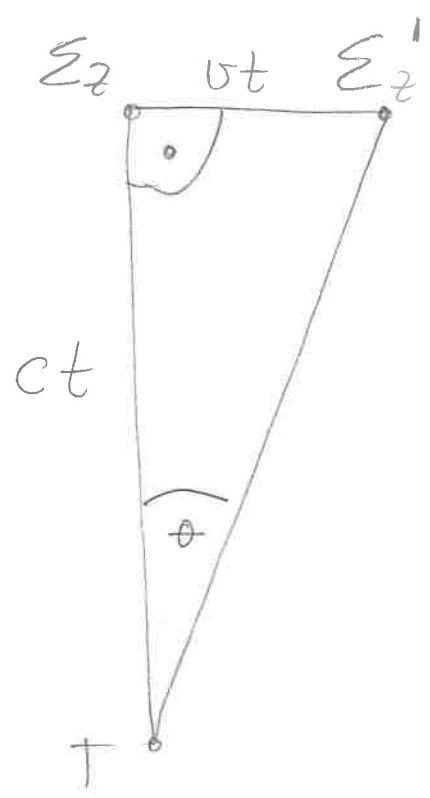
ABERACIJA SVETLOSTI NASTAJE JER JE  
BRZINA SVETLOSTI KONAČNA, SAMIM TIM  
SVETLOSTI JE POTREBNO NEKO KONAČNO VREME  
DA PROĐE KROZ TUBU TELESKOPA. TOFOM TOG  
VREMENA ~~SE~~ POSMATRAČ SE ~~PREĆE~~ USLED  
ROTACIJE I REVOLUCIJE



AKO BI NEKO GADAO VAGON ~~U~~ U POKRETU  
IZ LOKACIJE  $\Sigma$  U PRAVCU  $\odot$ , POGOUDIO  
BI TAČKU B KOJA SE NALAZI NA LEVOJ  
STRANI VAGONA. DO ISTOG REZULTATA  
BI DOŠLO AKO BI GADAO IZ LOKACIJE  
 $\Sigma'$  KROZ  $O'$ , A DA VAGON MIRUJE.

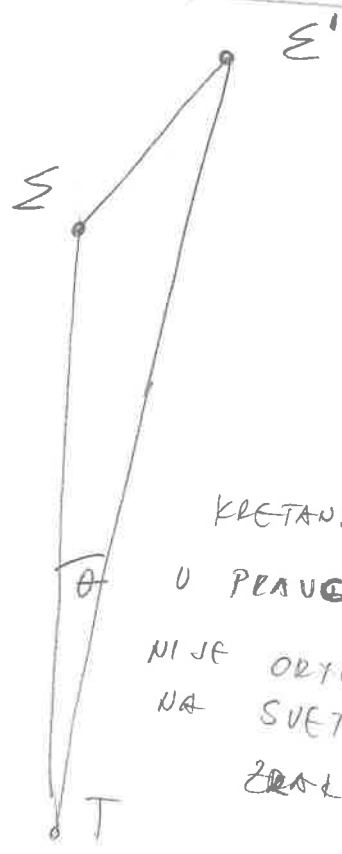


ZVEZDA U ZENITU



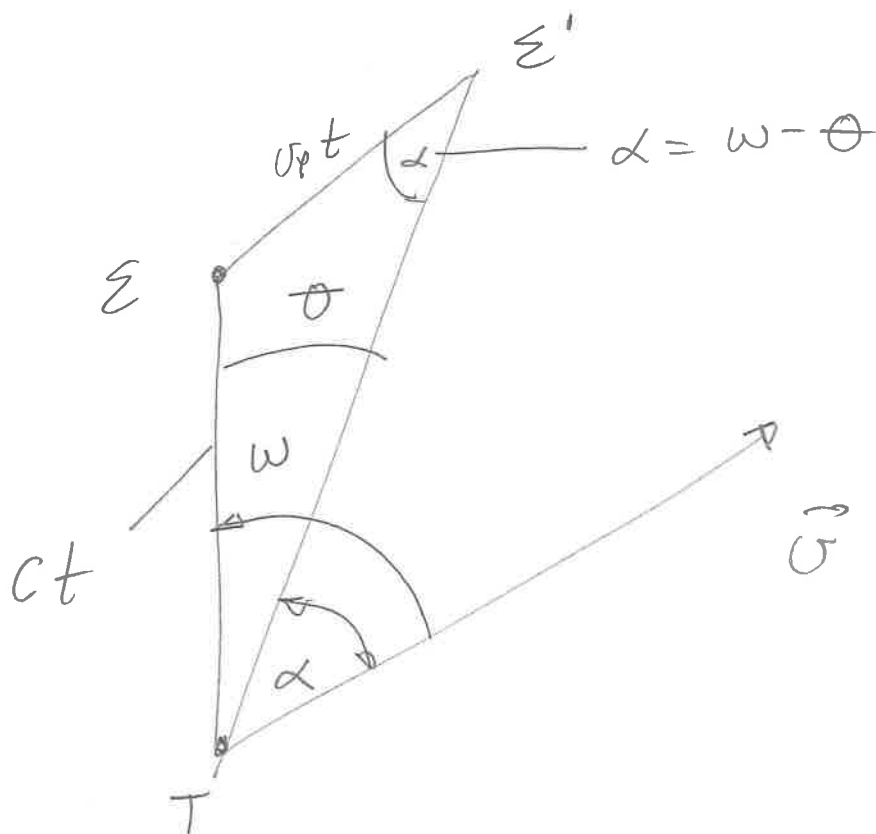
KRETANJE POSMATRAČA U  
PRAVCU ORTOGONALNOM NA  
SVETLOSNE ZRAKE

ZVEZDA NIJE U ZENITU



KRETANJE POSMATRAČA  
U PRAVCU KOJI  
NIJE ORTOGONALAN  
NA SVETLOSNE  
ZRAKE

ZVEZDA NIE U ZENITU



$$\Delta \Sigma \Sigma' T \quad \frac{\sin \theta}{v_p t} = \frac{\sin (\omega - \theta)}{ct}$$

$$\theta \approx \sin \theta = \frac{v_p}{c} \sin (\omega - \theta) \approx \frac{v_p}{c} \sin \omega$$

$$\theta'' = \theta(\text{rad}), \sin 1'' \approx \sin [\theta(\text{rad})] 1''^c$$

$$\Rightarrow \theta'' = \frac{v_p}{c \sin 1''} \sin \omega$$

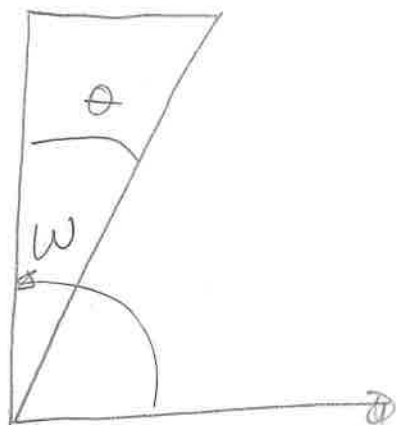
KONSTANTA ABERACIJE

$k_\varphi$  (ZA DATU  
GEOGRAFSKU  
ŠIRINU  $\varphi$ )

$$k_\varphi = \frac{v_p}{c \sin 1''} = \frac{v_e \cos \varphi}{c \sin 1''} = k' \cos \varphi$$

$k'$  - KONSTANTA DREVNE  
ABERACIJE NA EKVATORU

KADA JE ZVEZDA U ZENITU ONDA JE  
UGAO KOD  $\Sigma$  PRAV ( $W = 90^\circ$ )



SADA JE

$$\theta = \frac{v_e}{c \sin 1''} = k_\gamma = k \cos \varphi$$

ZA ZVEZDU U ZENITU ABERACIJSKI

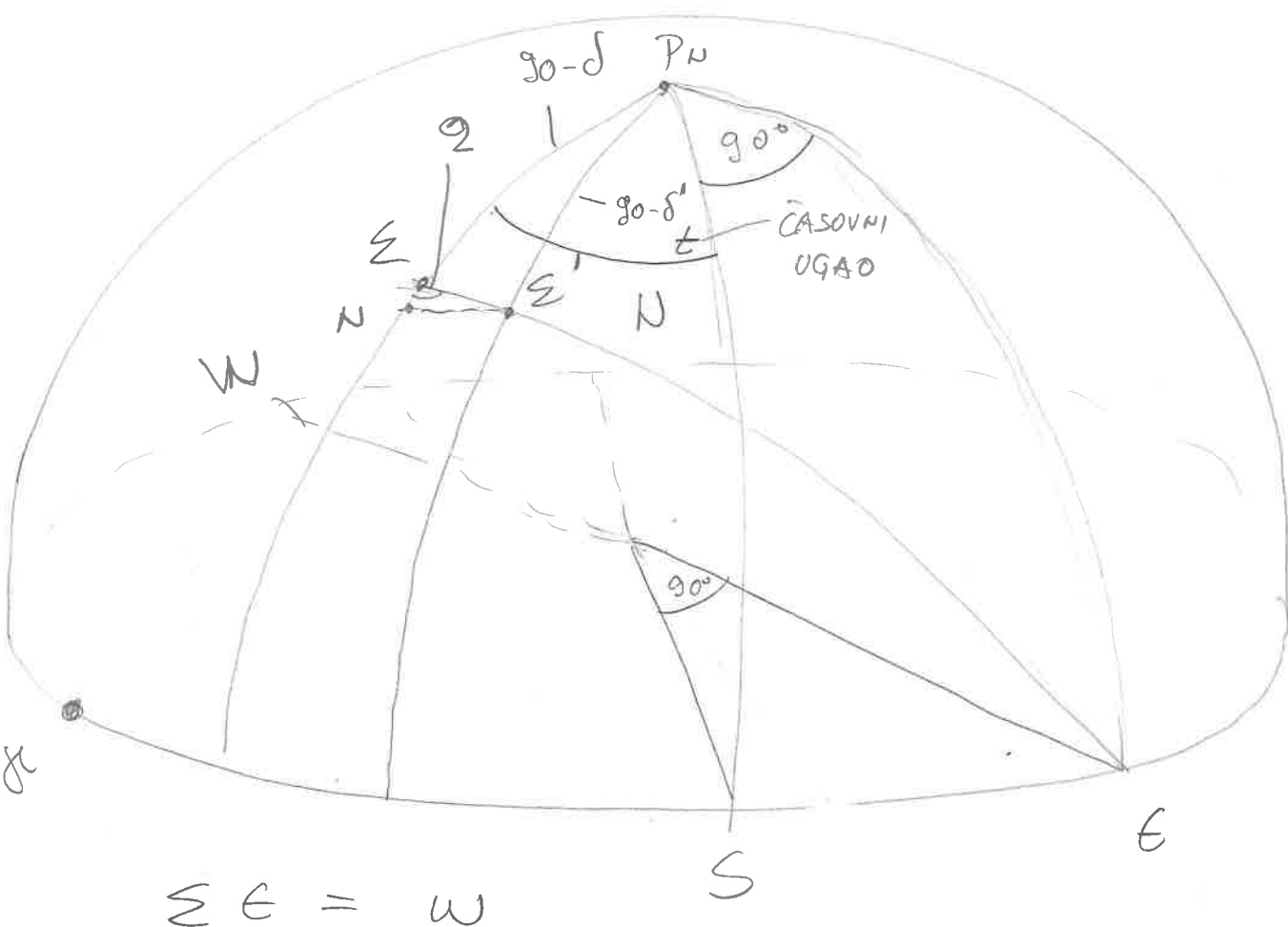
UGAO JE JEDNAK KONSTANTI ABERACIJE

$$k' = \frac{v_e}{c \sin 1''} \approx 32''$$

DAKLE, USLED DNEVNE ABERACIJE POLOZAJ  
ZVEZDE BIĆE POMEREN ~~KA~~ ISTOČNOJ TACNI  
HORIZONTA ZA MALI UGAO

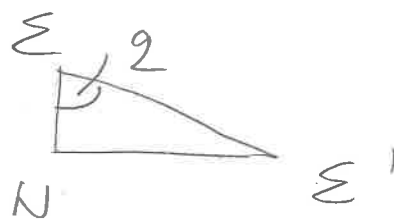
$$\Sigma \Sigma' = 32'' \cdot \cos \varphi \sin W$$

UTICAJ DNEVNE ABERACIJE NA  
EKVATORSKE KOORDINATE



$$\Sigma E = W$$

$\Delta \Sigma \Sigma' N$



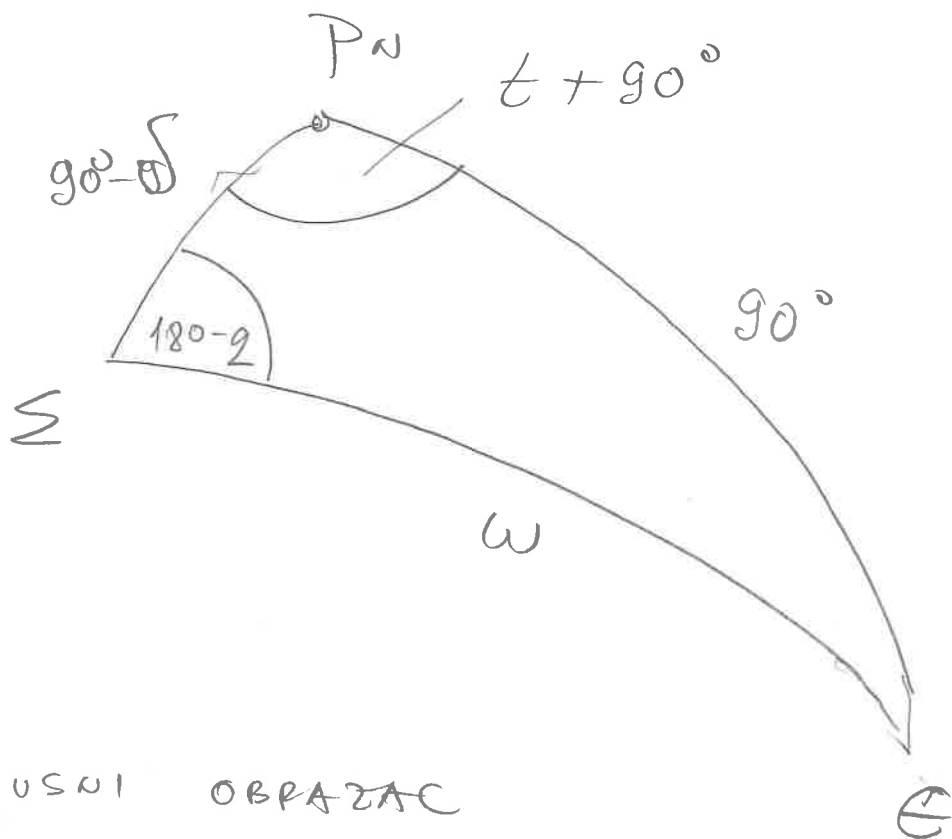
$$(1) \begin{cases} \delta' - \delta = \Sigma N = \Sigma \varepsilon' \cos \varphi = k \varphi \sin \omega \cos \varphi \\ (\alpha' - \alpha) \cos \delta = \Sigma' N = \Sigma \varepsilon' \sin \varphi = k \varphi \sin \omega \sin \varphi \end{cases}$$

$$\sin w \cos z \quad i \quad \sin w \sin z \quad \text{добавим}$$

12

$$\Delta \Sigma P_N \in$$

⑥



- SINUSNI OBRAZAC

$$\frac{\sin \omega}{\sin(t + 90^\circ)} = \frac{\sin 90^\circ}{\sin(180 - \gamma)} = \frac{1}{\sin(\gamma)}$$

$$\sin \omega \sin \gamma = \sin(t + 90^\circ) = \cos t$$

- SINUSNO - KOSINUSNI OBRAZAC

$$\sin \omega \cos(180 - \gamma) = \cancel{\cos 90^\circ} \sin(90 - \delta) = \sin 90^\circ \cos(90 - \delta),$$

$\cos(90 + t)$

$$-\sin \omega \cos \gamma = \sin \delta \sin t$$

KADA SE OVI IZRAZI UVRSTE U (1)  $\Rightarrow$

$$\alpha' - \alpha = k \varphi \sec \delta \cos t$$

$$\delta' - \delta = k \varphi \sin \delta \sin t$$

# GODIŠNJA ABERACIJA

Dešava se zbog zemljnog kretanja oko Sunca.

$$v \approx \frac{2\pi a}{T} \approx 30 \text{ km/s}$$

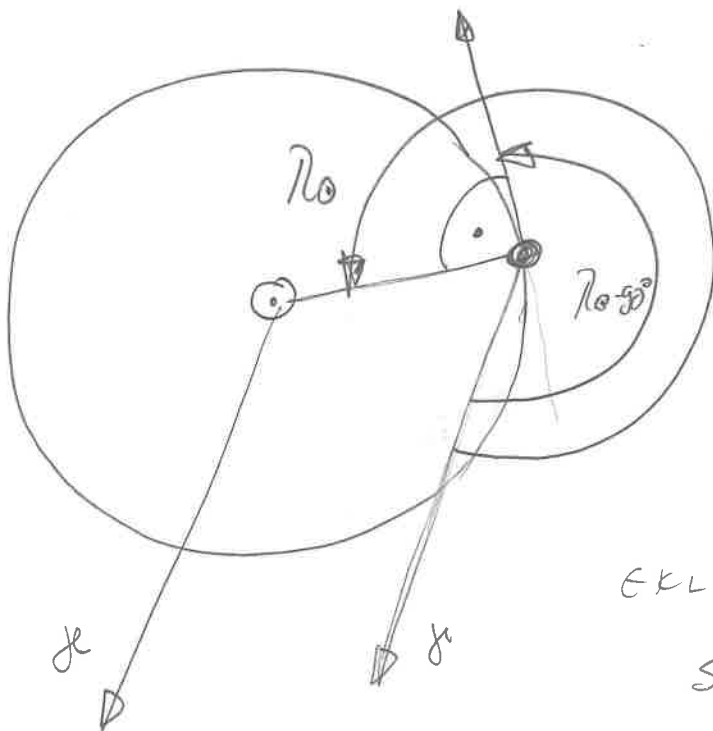
VELIKA POLUOSA — BRZINA ZEMLJINOG KRETANJA

T — SIDERIČKA GODINA

KONSTANTA ABERACIJE

$$k = \frac{v}{c \sin 1''} \approx 20'',5 \text{ OKO 40 PUTA VEĆA OD DNEVNE}$$

TREKUTNI APEKS



APEKS JE TAČKA

NA EKLIPTICI ČIJA

EKLIPTIČKA LONGITUDA

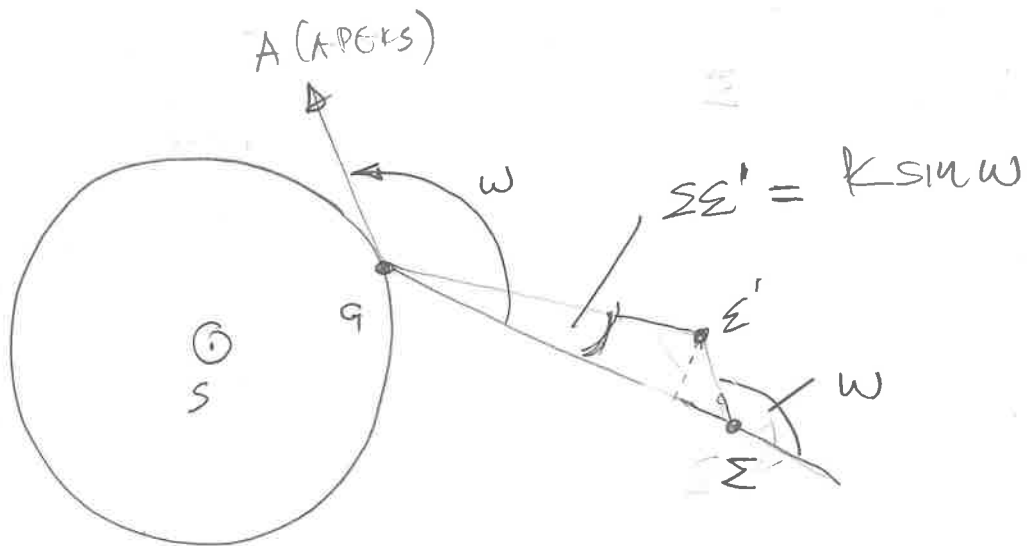
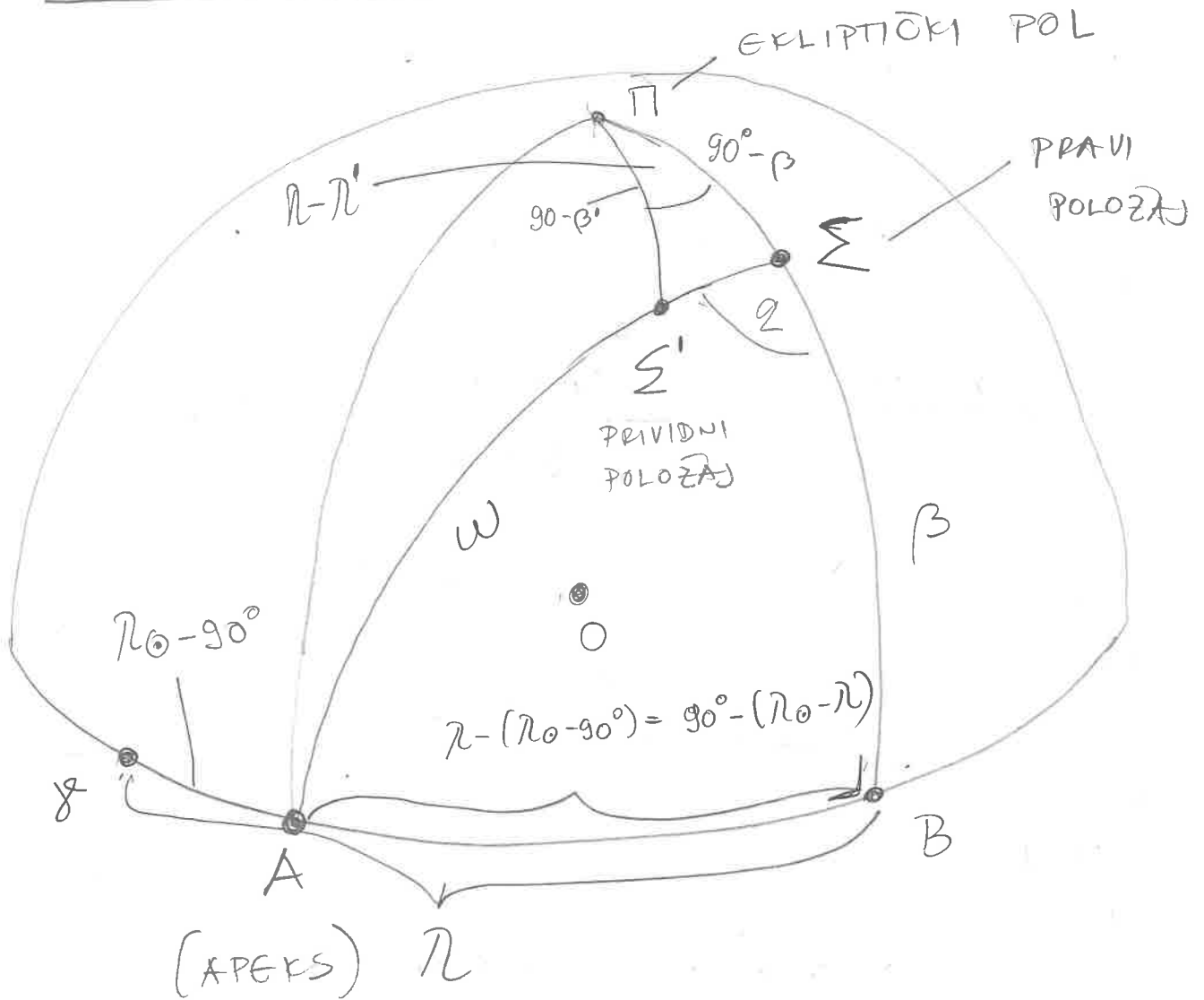
JE UVEK ZA

$90^\circ$  MANJA OD

EKLIPTIČKE LONGITUDE

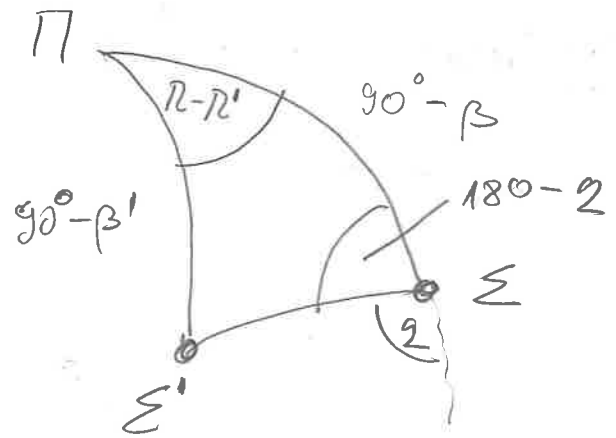
SUNCA

UTICAJ GODIŠNJE ABERACIJE NA  
EKLIPTIČKE KOORDINATE





$\Delta \Pi \Sigma \Sigma'$



SINUSNI OBRAZAC

$$\frac{\sin(\pi - \pi')}{\sin \Sigma \Sigma'} = \frac{\sin(180^\circ - 2)}{\sin(90^\circ - \beta')} = \frac{\sin 2}{\cos \beta'}$$

$$\sin(\pi - \pi') \cos \beta' = \sin \Sigma \Sigma' \sin 2$$

$\pi - \pi'$  JE MALI UGLO PA JE  $\sin(\pi - \pi') \approx \pi - \pi'$

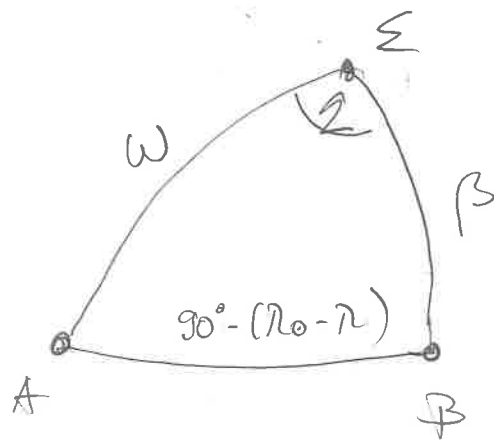
$$\cos \beta' \approx \cos \beta$$

$$\sin \Sigma \Sigma' \approx \Sigma \Sigma' = k \sin w$$

$$\Rightarrow (\pi - \pi') \cos \beta = k \sin w \sin 2 \quad (1)$$

NEPOZNATA VREDNOST  
NALAZIMO JE IZ PRAVOUGLOG  
SFERNOG TROUGLA  $A \Sigma B$

$\triangle A \Sigma B$



$$\sin \omega = \frac{\sin(90^\circ - (\lambda_0 - \lambda))}{\sin q} = \frac{\cos(\lambda_0 - \lambda)}{\sin q}$$

tj.  $\sin \omega \sin q = \cos(\lambda_0 - \lambda)$

ZAMENOM OVOG IZRAZA U (1)  $\Rightarrow$

$$\boxed{\lambda' - \lambda = -k \cos(\lambda_0 - \lambda) \sec \beta} \quad (2)$$

PROJENA EKLIPTIČKE LONGITUDE  
USLED GODIŠNJE ABERACIJE

UTICAJ NA EKLIPTIČKU LATITUDU SE  
DOBIJA IZ ISTA DVA TROUGLA

$\triangle \Pi \Sigma \Sigma'$

SINUSNO - KOSINUSNI OBRATAC

$$\sin \Sigma \Sigma' \cos(180^\circ - q) = \sin \beta' \cos \beta - \cos \beta' \sin \beta \cos(\lambda - \lambda')$$

POŠTO JE  $\lambda - \lambda' \approx 0 \Rightarrow \cos(\lambda - \lambda') \approx 1$

$$\Rightarrow \Sigma \Sigma' \cos(180^\circ - q) = \sin(\beta' - \beta) \approx \beta' - \beta$$

$$\beta' - \beta = -k \sin w \cos q$$

posto je

$$\cos(180 - q) = -\cos q$$

tj.

$$\beta - \beta' = \underbrace{k \sin w \cos q}_{\text{NEPOZNATA VREDNOST KOSU NALAZIMO IZ PRAVOUGLOG SFERNOG TROUGLA AEB}} \quad (3)$$

NEPOZNATA VREDNOST

KOSU NALAZIMO IZ

PRAVOUGLOG SFERNOG TROUGLA

AEB

SINUSNO - KOSINUSNI OBRAZAC

$$\sin w \cos q = \underbrace{\cos(90^\circ - (\lambda_0 - \lambda))}_{\sin(\lambda_0 - \lambda)} \sin \beta - \cancel{\sin(90^\circ - (\lambda_0 - \lambda)) \cos \beta \cdot \cos 90^\circ}$$

$$\sin w \cos q = \sin(\lambda_0 - \lambda) \sin \beta$$

UVRSTIMO OVO U (3)

$$\boxed{\beta' - \beta = -k \sin(\lambda_0 - \lambda) \sin \beta} \quad (4)$$

PROMENA

GEOMETRIČKE

LATITUDE

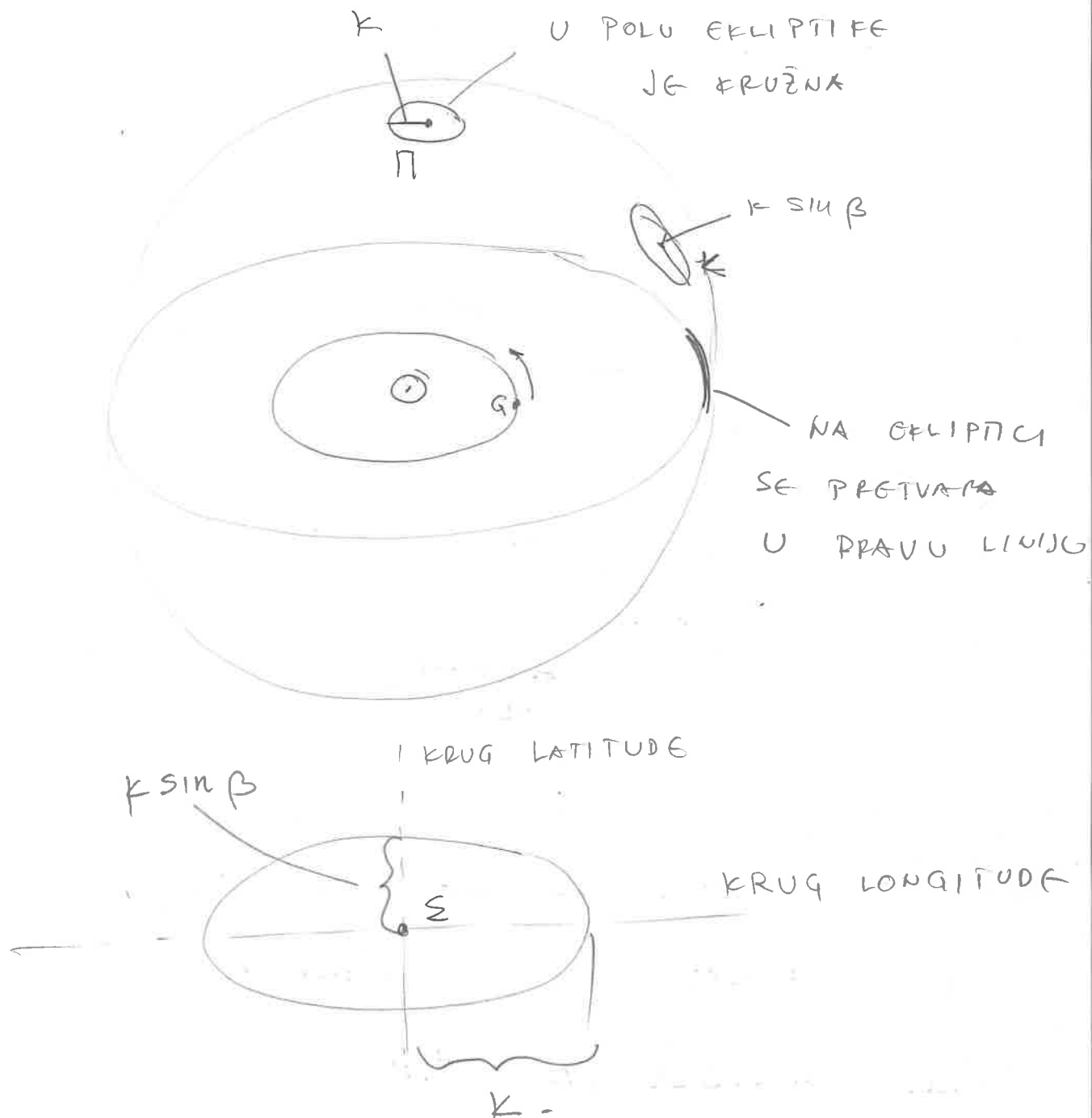
USLGO

GOĐIŠNJE

ABERACIJE



# ELIPSA GODIŠNJE ABERACIJE

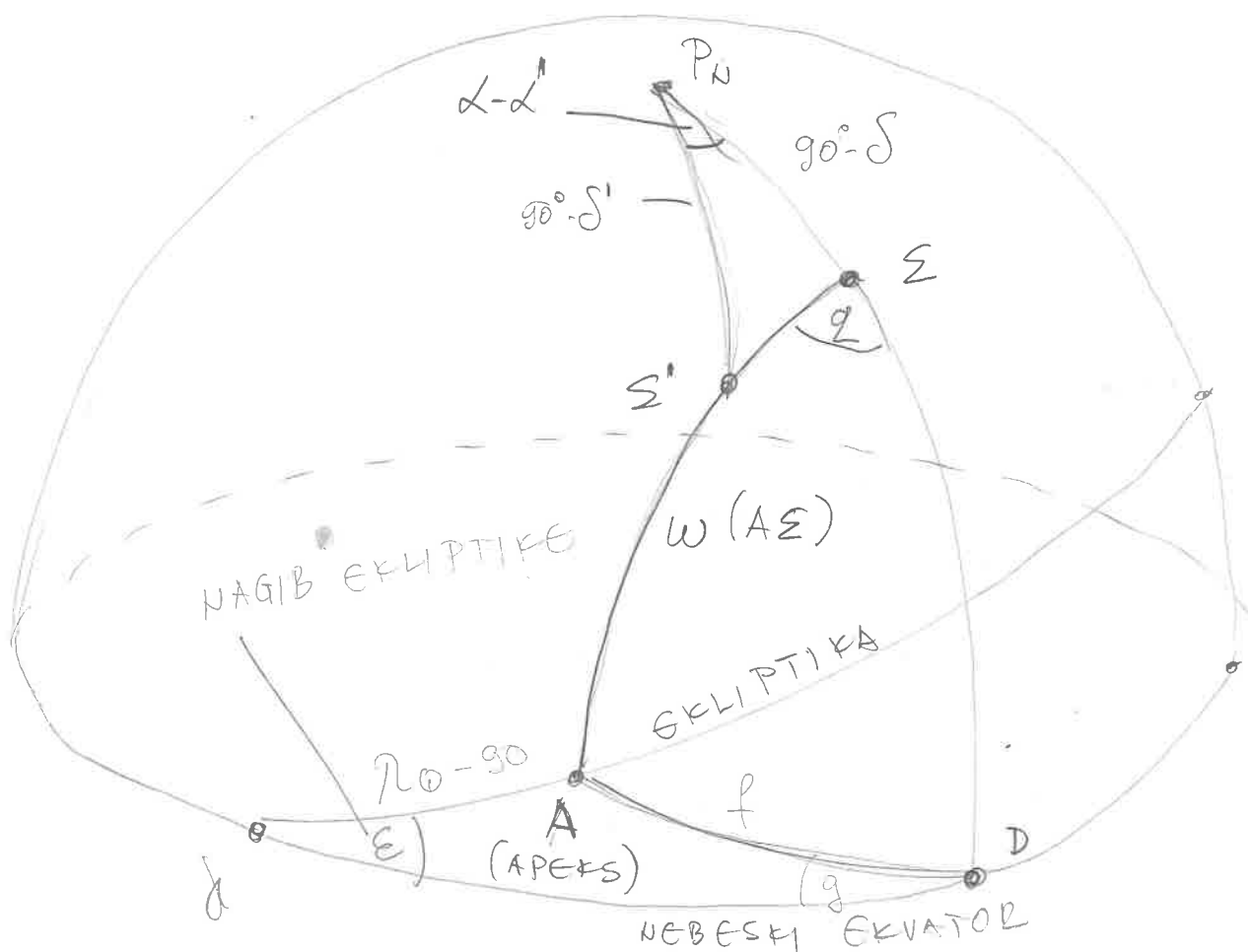


ELIPSA GODIŠNJE ABERACIJE

TREBA UOČITI DA JE PROMENA PRIVIDNOG  
POLOŽAJA USLED GODIŠNJE ABERACIJE  
MNOGO VEĆA NEGO USLED GODIŠNJE PARALAKSE.

# UTICAJ GODIŠNJE ABERACIJE NA EKVATORSKE KOORDINATE

$$\alpha' - \alpha \quad \delta' - \delta$$



SFERNI TRIUGAO  $\triangle P_N \Sigma \Sigma'$

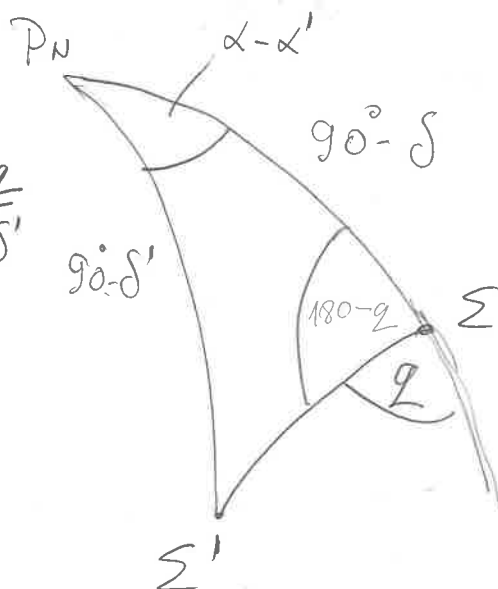
SINUSNI OBRATKE

$$\frac{\sin(\alpha - \alpha')}{\sin(\Sigma' \Sigma)} = \frac{\sin(180 - z)}{\sin(90 - \delta')} = \frac{\sin z}{\cos \delta'}$$

$$\sin(\alpha - \alpha') \approx \alpha - \alpha'$$

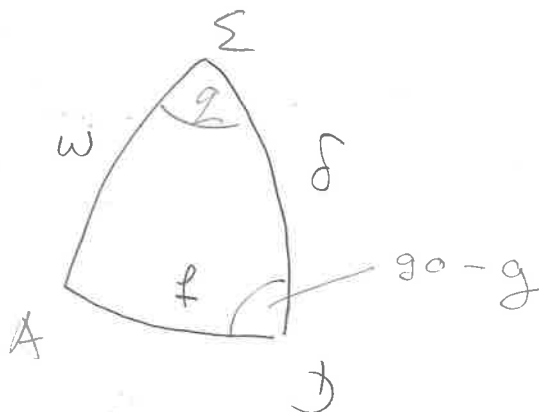
$$\cos \delta' \approx \cos \delta$$

$$\Sigma' \Sigma = k \sin W$$



$$\Rightarrow (\alpha - \alpha') \cos \delta = \underbrace{k \sin w \sin q}_{\text{NEPOZNATO}} \quad (1)$$

$\triangle \Sigma DA$



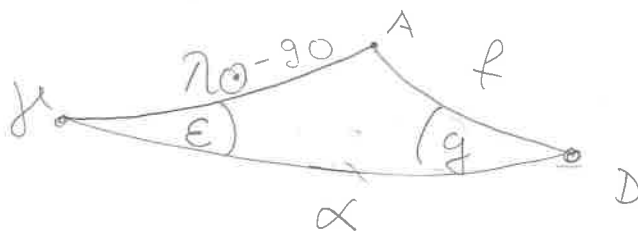
SINUSNA FORMULA

$$\frac{\sin w}{\sin(90^\circ - g)} = \frac{\sin f}{\sin q}$$

$$\text{tj.} \quad \sin w \sin q = \underbrace{\sin f \cos g}_{\text{NEPOZNATO}} \quad (2)$$

$\triangle AD\epsilon$

SINUSNO - KOSINUSNA FORMULA



$$\sin f \cos g = \cos(\lambda_0 - 90^\circ) \sin \alpha - \sin(\lambda_0 - 90^\circ) \cos \alpha \cos E$$

tj.

$$\sin f \cos g = \sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos E$$

ZAMENOM U (2) TJ. U (1)  $\Rightarrow$

$$\alpha' - \alpha = -k \sec \delta (\sin \lambda_0 \sin \alpha + \cos \lambda_0 \cos \alpha \cos E)$$

PROJENA REKTASCENZIJE USLED GODIŠNJE ABERACIJE

ZA UTICAJ NA  $\delta$  PRIMENJUJE SE  
SINUSNO - KOSINUSNI OBRAZAC NA

$$\Delta P_N \Sigma E'$$

$$\underbrace{\sin \Sigma \Sigma'}_{\approx \Sigma \Sigma'} \underbrace{\cos(180^\circ - \varrho)}_{-\cos \varrho} = \underbrace{\sin(\delta' - \delta)}_{\approx \delta' - \delta}$$

$$\Rightarrow \delta' - \delta = - \underbrace{\sin W \cos \varrho}_{\text{NEPOZNATO}} \quad (3)$$

NEPOZNATO  
DOBIVA SE IZ  $\Delta \Sigma DA$  POMOĆU  
SINUSNO - KOSINUSNOG OBRAZCA

$$\sin W \cos \varrho = \cos \varphi \sin \delta - \cos \delta \sin \varphi \cos(90^\circ - \varrho)$$

tj.

$$\sin W \cos \varrho = \underbrace{\cos \varphi \sin \delta}_{\text{NEPOZNATO}} - \underbrace{\cos \delta \sin \varphi \sin \varrho}_{\text{NEPOZNATO}} \quad (4)$$

SADA PONOVO KORISTIMO TROUGAO  $\Delta AD \gamma$

KOSINUSNI OBRAZAC

$$\cos \varphi = \cos \alpha \cos(\lambda_0 - 90^\circ) + \sin \alpha \sin(\lambda_0 - 90^\circ) \cos E$$

tj.

$$\cos \varphi = \cos \alpha \sin \lambda_0 - \sin \alpha \cos \lambda_0 \cos E \quad (5)$$



## SINUSNI OBRATAC

$$\frac{\sin f}{\sin \epsilon} = \frac{\sin(\lambda_0 - \alpha_0)}{\sin g} = - \frac{\cos \lambda_0}{\sin g}$$

tj.

$$\sin f \sin g = - \cos \lambda_0 \sin \epsilon \quad (6)$$

ZAMENOM (5) I (6) U (4), PA POTOM U (3)

$$\delta' - \delta = -k \left[ \sin \lambda_0 \cos \alpha \sin \delta + \cos \lambda_0 \cos \epsilon (\operatorname{tg} \epsilon \cos \delta - \sin \alpha \sin \delta) \right]$$

PROMENA DEKLINACIJE USLED GODIŠNJE ABERACIJE

IZRAZI ZA  $\alpha' - \alpha$  I  $\delta' - \delta$  MOGU SE PRIKAZATI U JEDNOSTAVNIJEM OBLIKU

$$\alpha' - \alpha = Cc + Dd$$

$$\delta' - \delta = Cc' + Dd'$$

$$\left. \begin{aligned} C &= -k \cos \lambda_0 \cos \epsilon \\ D &= -k \sin \lambda_0 \end{aligned} \right\} \begin{array}{l} \text{BESELOVI} \\ \text{ISTI SU ZA SVE ZVEZDE} \end{array}$$

$$C = \frac{1}{15} \cos \alpha \sec \delta, \quad C' = \operatorname{tg} \epsilon \cos \delta - \sin \alpha \sin \delta$$

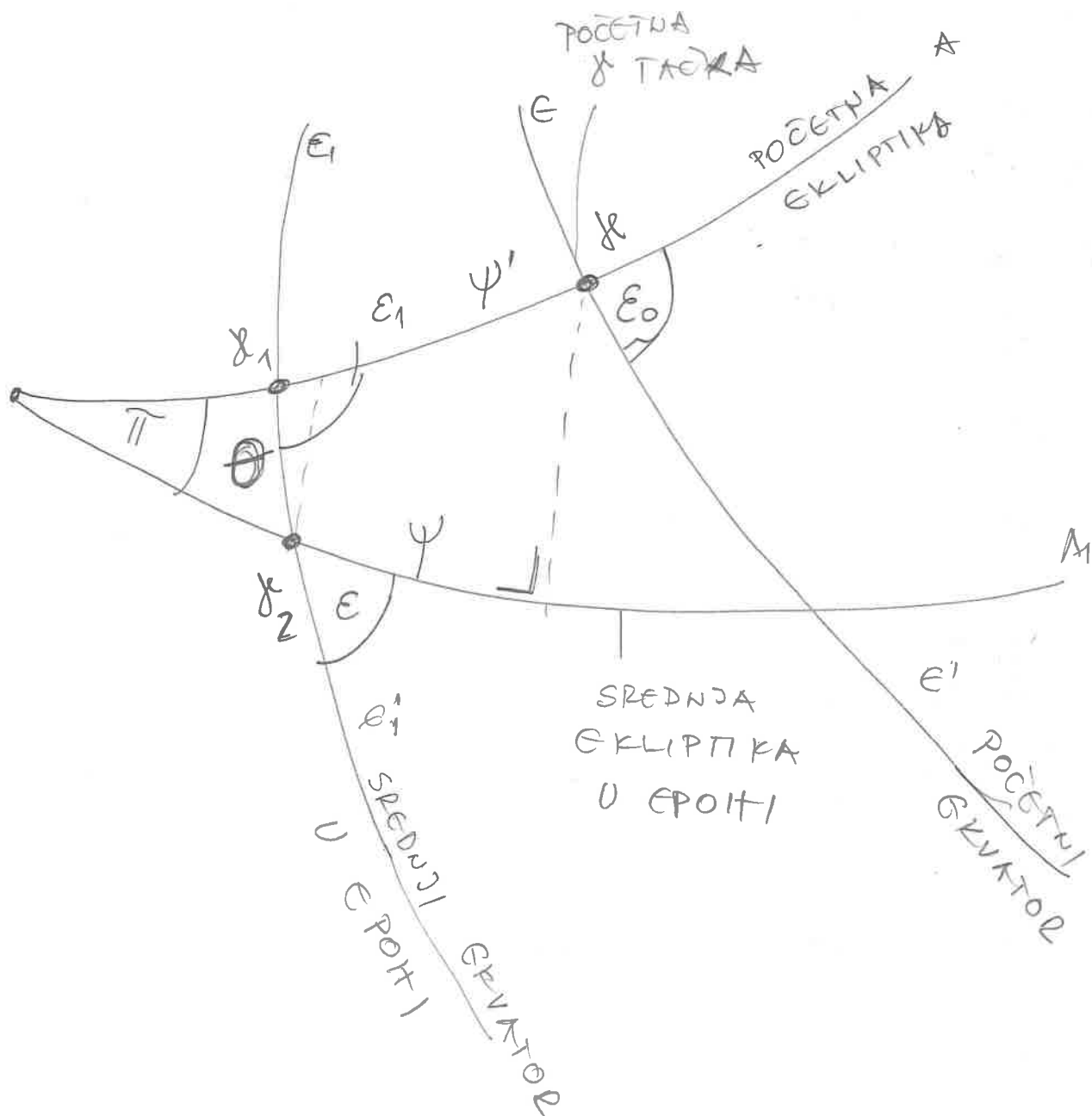
$$d = \frac{1}{15} \sin \alpha \sec \delta, \quad d' = \cos \alpha \sin \delta$$

FAKTOR  $\frac{1}{15}$  SLUŽI ZA PRETVARANJE IZ UGLOVNIH U DRAŽOVNE JEDINICE

# PRECESIJA

- HIPARH JE OTKRIO DA TROPSKA GODINA  
TRAJE KRAĆE OD SIDERIČKE, ŠTO JE  
POSLEDICA PRECESIJE.

- LUNI - SOLARNA PRECESIJA } + OPŠTA  
- PLANETSKA PRECESIJA } PRECESIJA



$\psi'$  - LUNI - SOLARNA PRECESIJA

$\gamma \longrightarrow \gamma_1$

$\epsilon$  - SREDNJI NAGIB EKLIPTIKE U EPOCH

$\theta$  - PLANETSKA PRECESIJA

$\gamma_2$  - SREDNJA  $\gamma$  TAČKA

SREDNJOJ EKLIPTICI I SREDNJOJ  $\gamma$  TAČKI  
ODGOVARAJU SREDNJI POLOVI I SREDNJI POLOŽAJI  
NEBESKIH TELA,

LUNI - SOLARNA PRECESIJA OKO  $50'',37$  GODIŠNJE  
PLANETSKA PRECESIJA OKO  $0'',13$  (U DIREKTNOM  
SMERU)

OPŠTA PRECESIJA

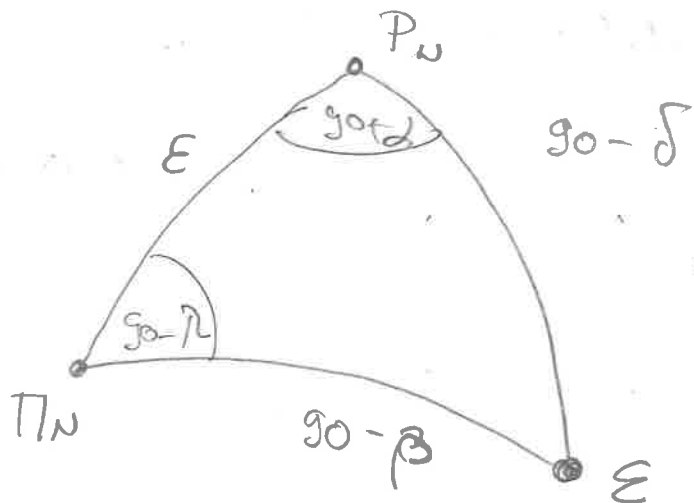
$$\psi = \underbrace{\psi'}_{\substack{\text{LUNI} \\ \text{SOLARNA}}} - \underbrace{\theta}_{\text{PLANETSKA}} \cos \epsilon_1$$

NAGIB EKVATORA  
U ODNOSU NA  
POČETNU  
EKLIPTIKU



# UTICAJ LUNI-SOLARNE PRECESIJE NA EKVATORSKE KOORDINATE

$$\triangle \Pi_N P_N \Sigma$$



KOSINUSNI OBRAZAC

$$\cos(90 - \delta) = \cos \epsilon \cos(90 - \beta) + \sin \epsilon \sin(90 - \beta) \cos(90 - \lambda)$$

$$\sin \delta = \cos \epsilon \sin \beta + \sin \epsilon \cos \beta \sin \lambda \quad (1)$$

SINUSNI OBRAZAC

$$\sin(90 - \beta) \sin(90 - \lambda) = \sin(90 - \delta) \sin(90 + \alpha)$$

$$\cos \beta \cos \lambda = \cos \delta \cos \alpha \quad (2)$$

DIFFERENCIRAMO (1) UZIMAJUĆI U OBZIR DA  
JE  $\beta = \text{const}$ ,  $\lambda \neq \text{const}$

$$\cos \delta d\delta = \sin \epsilon \underline{\cos \beta \cos \lambda} d\lambda \quad (3)$$

$$(2) \rightarrow (3) \Rightarrow$$

$$\cancel{\cos \delta} d\delta = \sin \varepsilon \cancel{\cos \delta} \cos \alpha d\mathcal{R}$$

$$d\delta = \sin \varepsilon \cos \alpha \underbrace{d\mathcal{R}}_{\mathcal{J}\Psi}$$

$$\boxed{d\delta = \mathcal{J}\Psi \sin \varepsilon \cos \alpha}$$

DIFFERENCIUJAMO (2) ( $\beta = \text{const}$ )

$$\sin \delta \cos \alpha d\delta + \cos \delta \sin \alpha d\alpha = \underline{\cos \beta \sin \mathcal{R} d\mathcal{R}} \quad (4)$$

$$\Delta \Pi_N \Pi_N \Sigma$$

SINUSNO - KOSINUSNI OBRATAC

$$\begin{aligned} \sin(90-\beta) \cos(90-\mathcal{R}) &= \cos(90-\delta) \sin \varepsilon \\ &\quad - \sin(90-\delta) \cos \varepsilon \underbrace{\cos(90+\alpha)}_{-\sin \alpha} \end{aligned}$$

$$\underline{\cos \beta \sin \mathcal{R}} = \sin \delta \sin \varepsilon + \cos \delta \cos \varepsilon \sin \alpha \quad (5)$$

$$(5) \rightarrow (4) \Rightarrow$$

$$\cos \delta \sin \alpha d\alpha = \underline{\cos \beta \sin \alpha d\alpha} - \sin \delta \cos \alpha d\delta \quad (4)$$

$$\begin{aligned} \cos \delta \sin \alpha d\alpha &= (\sin \delta \sin \varepsilon + \cos \delta \cos \varepsilon \sin \alpha) d\alpha \\ &\quad - \sin \delta \cos \alpha d\delta \end{aligned}$$

$$\boxed{d\delta = \psi \delta \sin \varepsilon \cos \alpha \quad d\alpha = \psi \delta}$$

$\Rightarrow$

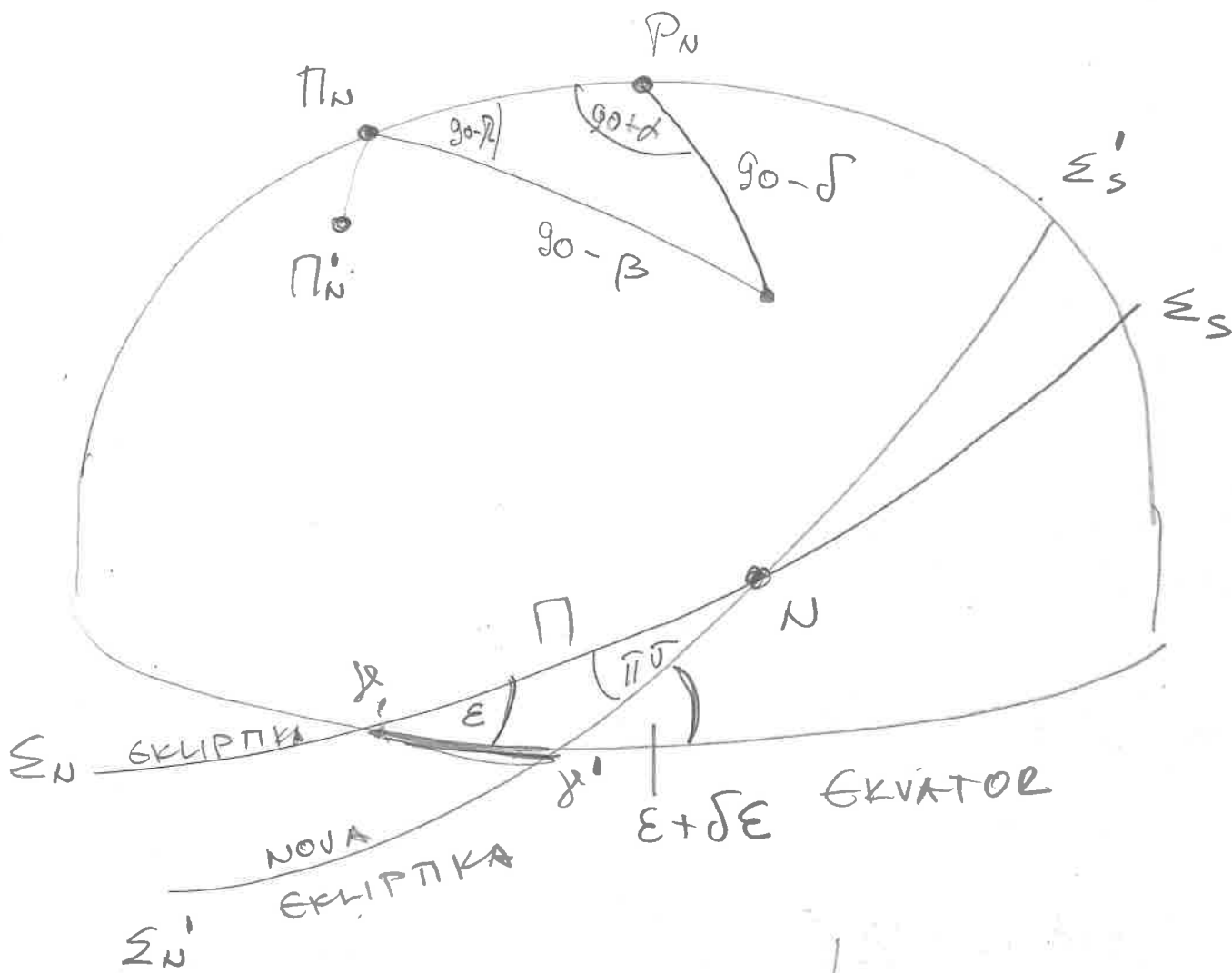
$$\begin{aligned} \cos \delta \sin \alpha d\alpha &= (\sin \delta \sin \varepsilon + \cos \delta \cos \varepsilon \sin \alpha) \psi \delta - \\ &\quad - \sin \delta \cos^2 \alpha \sin \varepsilon \psi \delta \end{aligned}$$

$$\cos \delta \sin \alpha d\alpha = \psi \delta (\sin \delta \sin \varepsilon + \cos \delta \cos \varepsilon \sin \alpha - \sin \delta \sin \varepsilon \cos^2 \alpha)$$

$$\cancel{\cos \delta \sin \alpha d\alpha} = \psi \delta (\cancel{\sin \delta \sin \varepsilon} + \cos \delta \cos \varepsilon \sin \alpha - \cancel{\sin \delta \sin \varepsilon \cos^2 \alpha})$$

$$\Rightarrow \boxed{d\alpha = \psi \delta (\cos \varepsilon + \tan \delta \sin \varepsilon \sin \alpha)}$$

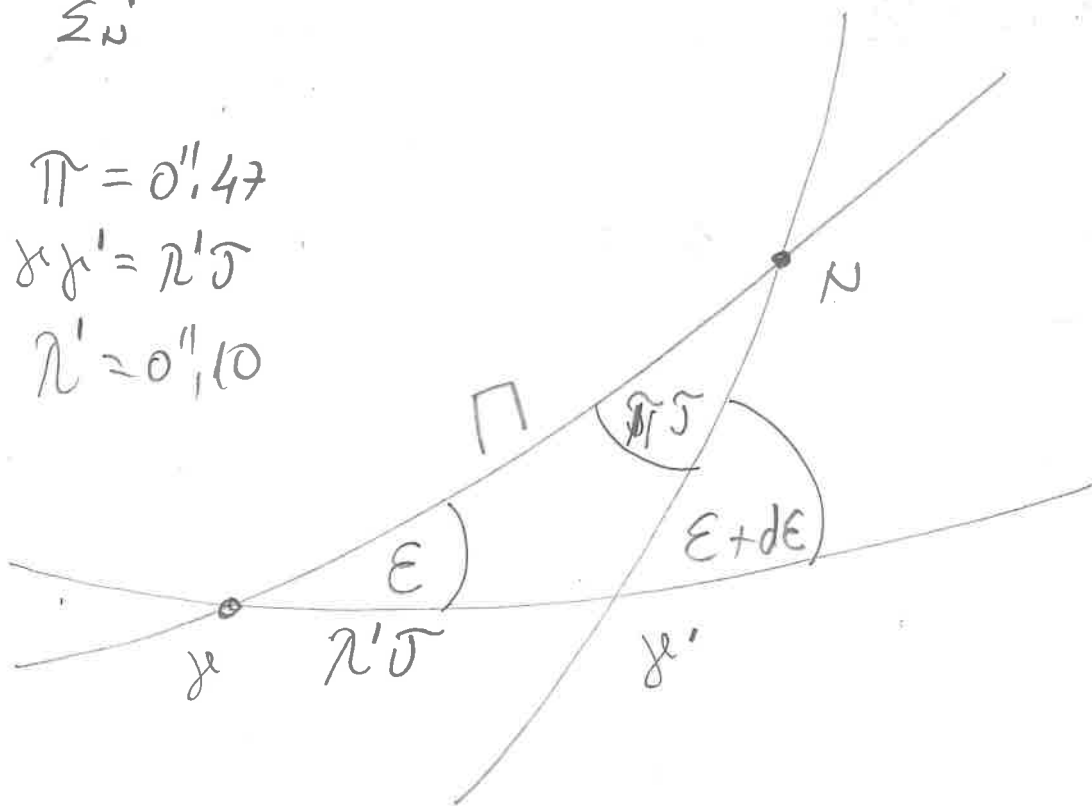
## PLANETSKA PRECESIJA



$$\pi = 0.47$$

$$\delta\delta' = \mathcal{R}'\mathcal{J}$$

$$\lambda' \approx 0'',10$$





UTICAJ PLANETSKOG PRECESIJE

NA EKVATORSKE KOORDINATE

$$d\alpha = -\alpha' \delta$$

$$d\delta = 0$$

UTICAJ PLANETSKOG PRECESIJE

NA EKLIPTIČKE KOORDINATE

$\triangle \gamma \gamma' N$

SINOSNI OBRZAC

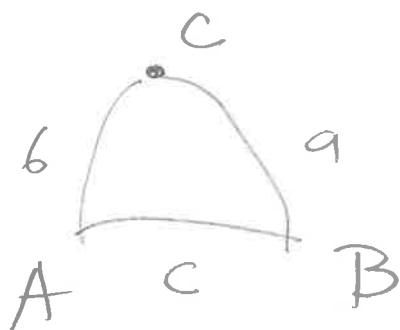
$$\sin \Pi \sin \Pi \delta = \sin(180 - (\epsilon + d\epsilon)) \cdot \sin \alpha' \delta$$

$$\sin \Pi \underbrace{\sin \Pi \delta}_{\approx \Pi \delta} = \underbrace{\sin(\epsilon + d\epsilon)}_{\approx \sin \epsilon} \cdot \underbrace{\sin \alpha' \delta}_{\approx \alpha' \delta}$$

$$\cancel{\Pi \delta} \sin \Pi = \cancel{\alpha' \delta} \sin \epsilon$$

$$\boxed{\alpha' = \Pi \sin \Pi \operatorname{cosec} \epsilon}$$

$$\cos a \cos C = \sin a \operatorname{ctg} b - \sin C \operatorname{ctg} B$$



$$\cos \rho' \sigma \cos \epsilon = \sin \rho' \sigma \operatorname{ctg} \Pi - \sin \epsilon \operatorname{ctg} (180 - (\epsilon + d\epsilon))$$

$$\cos \rho' \sigma \cos \epsilon = \sin \rho' \sigma \operatorname{ctg} \Pi + \sin \epsilon \operatorname{ctg} (\epsilon + d\epsilon) \cdot \sin(\epsilon + d\epsilon)$$

$$\underbrace{\sin(\epsilon + d\epsilon)}_{\approx 1} \cdot \underbrace{\cos \rho' \sigma \cos \epsilon}_{\approx 1} - \sin \epsilon \cos(\epsilon + d\epsilon) = \underbrace{\sin \rho' \sigma \operatorname{ctg} \Pi}_{\rho' \sigma} \cdot \underbrace{\sin(\epsilon + d\epsilon)}_{\approx \sin \epsilon}$$

$$\sin(\epsilon + d\epsilon) \cos \epsilon - \cos(\epsilon + d\epsilon) \sin \epsilon = \rho' \sigma \operatorname{ctg} \Pi \sin \epsilon$$

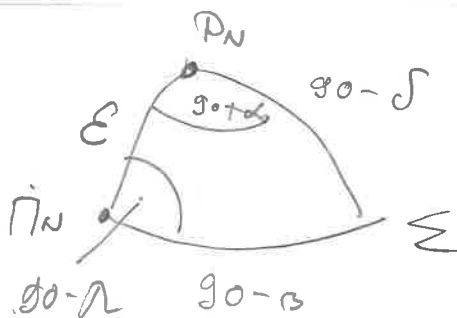
$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\sin(\epsilon + d\epsilon - \epsilon) = \rho' \sigma \operatorname{ctg} \Pi \sin \epsilon$$

$$\sin d\epsilon = \rho' \sigma \operatorname{ctg} \Pi \sin \epsilon$$

$$d\epsilon = \rho' \sigma \operatorname{ctg} \Pi \sin \epsilon$$

$$\triangle P_N \Pi_N \Sigma$$



KOSINUSNI OBRAZAC

$$\cos(90 - \beta) = \cos \epsilon \cos(90 - \delta) + \sin \epsilon \sin(90 - \delta) \cos(90 + \alpha)$$

$$\sin \beta = \cos \epsilon \sin \delta - \sin \epsilon \cos \delta \sin \alpha$$

DIFFERENCIRAMO OVAJ IZRAZ UZIMAJUĆI  
U OBZIR DA JE  $\beta \neq \text{const}$ ,  $\epsilon \neq \text{const}$   
 $\alpha \neq \text{const}$ ,  $\delta = \text{const}$ .

$$\cos \beta d\beta = -\sin \epsilon \sin \delta d\epsilon - \cos \epsilon \cos \delta \sin \alpha d\epsilon \\ - \sin \epsilon \cos \delta \cos \alpha d\alpha$$

$$\cos \beta d\beta = - \left( \sin \epsilon \sin \delta + \cos \epsilon \cos \delta \sin \alpha \right) d\epsilon \quad (6) \\ - \sin \epsilon \cos \delta \cos \alpha d\alpha$$

SINUSNO - KOSINUSNI OBRAZAC

$$\sin(90 - \beta) \cos(90 - \lambda) = \cos(90 - \epsilon) \sin \delta - \sin(90 - \delta) \cos \epsilon \cos(90 - \alpha) \\ \cos \beta \sin \lambda = \sin \delta \sin \epsilon + \cos \delta \cos \epsilon \sin \alpha \quad (7)$$

# SINUSNI OBRAZAC

$$\sin(90-\beta) \sin(90-\pi) = \sin(90-\delta) \sin(90+\alpha)$$

$$\cos \beta \cos \pi = \cos \delta \cos \alpha \quad (8)$$

$$(7)(8) \longrightarrow (6)$$

$$\cancel{\cos \beta} d\beta = -\cancel{\cos \beta} \sin \pi d\epsilon - \sin \epsilon \cancel{\cos \beta} \cos \pi d\alpha$$

$$d\beta = -\sin \pi d\epsilon - \sin \epsilon \cos \pi d\alpha$$

$$d\beta = -\sin \pi \underbrace{\pi' \delta \sin \epsilon \tan \pi}_{d\epsilon} + \cancel{\sin \epsilon \cos \pi \pi' \delta \sin \pi \operatorname{cosec} \epsilon}$$

$$d\alpha = -\pi' \delta =$$

$$-\pi' \delta \frac{\sin \pi}{\operatorname{cosec} \epsilon}$$

$$d\beta = -\sin \pi \cancel{\sin \epsilon} \tan \pi \cdot \underbrace{\pi' \delta \sin \pi \operatorname{cosec} \epsilon}_{\pi' \delta} + \pi' \delta \cos \pi \sin \pi$$

$$d\beta = -\sin \pi \pi' \delta \cos \pi + \pi' \delta \cos \pi \sin \pi$$

$$d\beta = -\pi' \delta (\sin \pi \cos \pi - \cos \pi \sin \pi)$$

$$d\beta = \pi' \delta (\sin \pi \cos \pi - \cos \pi \sin \pi)$$

$$d\beta = \pi' \delta \sin(\pi - \pi)$$

### SINUSNI OBRABAC

$$\sin(90 - \beta) \sin(90 - \lambda) = \sin(90 - \delta) \sin(90 + \alpha)$$

$$\cos \beta \cos \lambda = \cos \delta \cos \alpha \quad \left( \begin{array}{l} \text{DIFFERENCIRAMO} \\ \delta = \text{CONST} \end{array} \right)$$

$$-\sin \beta \cos \lambda d\beta - \cos \beta \sin \lambda d\lambda = -\cos \delta \sin \alpha d\alpha$$

$$\cos \beta \sin \lambda d\lambda = \underline{\cos \delta \sin \alpha d\alpha} - \sin \beta \cos \lambda d\beta \quad (9)$$

### SINUSNO - KOSINUSNI OBRABAC

$$\sin(90 - \delta) \cos(90 + \alpha) = \cos(90 - \beta) \sin \epsilon - \sin(90 - \beta) (\cos \epsilon \cos(90 - \lambda))$$

$$-\cos \delta \sin \alpha = \sin \beta \sin \epsilon - \cos \beta \cos \epsilon \sin \lambda$$

$$\underline{\cos \delta \sin \alpha} = -\sin \beta \sin \epsilon + \cos \beta \cos \epsilon \sin \lambda \quad (10)$$

$$(10) \longrightarrow (9)$$

$$\cos \beta \sin \lambda d\lambda = -(\sin \beta \sin \epsilon - \cos \beta \cos \epsilon \sin \lambda) d\lambda - \sin \beta \cos \lambda d\beta$$

$$d\alpha = -\lambda' \delta = -\pi \delta \frac{\sin \pi}{\sin \epsilon}$$

$$\cos \beta \sin \lambda d\lambda = -(\sin \beta \sin \epsilon - \cos \beta \cos \epsilon \sin \lambda) \cdot \overbrace{\left( -\pi \delta \frac{\sin \pi}{\sin \epsilon} \right)}^{d\alpha} - \sin \beta \cos \lambda \pi \delta \sin(\pi - \lambda) d\beta$$

$$\cos \beta \sin \lambda d\lambda = \pi \delta \left[ \sin \beta \sin \Pi - \cos \beta \sin \lambda \operatorname{ctg} \epsilon \cdot \sin \Pi \right. \\ \left. - \sin \beta \cos^2 \lambda \sin \Pi + \sin \beta \cos \lambda \cos \Pi \sin \lambda \right]$$

$$\cos \beta \sin \lambda d\lambda = \pi \delta \left[ \sin \beta \sin \Pi \underbrace{(1 - \cos^2 \lambda)}_{\sin^2 \lambda} - \cos \beta \sin \lambda \operatorname{ctg} \epsilon \sin \Pi \right. \\ \left. + \sin \beta \cos \lambda \cos \Pi \sin \lambda \right] / \cos \beta \sin \lambda$$

$$d\lambda = \pi \delta \left[ \operatorname{tg} \beta \sin \Pi \sin \lambda - \operatorname{ctg} \epsilon \sin \Pi + \operatorname{tg} \beta \cos \lambda \cos \Pi \right]$$

$$d\lambda = \pi \delta \left[ \operatorname{tg} \beta \underbrace{(\sin \Pi \sin \lambda + \cos \Pi \cos \lambda)}_{\cos(\Pi - \lambda)} - \operatorname{ctg} \epsilon \sin \Pi \right]$$

$$d\lambda = - \underbrace{\left( \pi \delta \frac{\cos \epsilon}{\sin \epsilon} \cdot \sin \Pi \right)}_{\lambda' \delta} + \pi \delta \operatorname{tg} \beta \cos(\Pi - \lambda)$$

$$d\lambda = -\lambda' \delta \cos \epsilon + \pi \delta \operatorname{tg} \beta \cos(\Pi - \lambda)$$

OPŠTA PRECESIJA

LS + PL

Ekvatorske koordinate

$$d\alpha = d\alpha_{LS} + d\alpha_{PL}$$

$$d\delta = d\delta_{LS} + d\delta_{PL}$$

$$d\alpha = -\lambda' \delta + \psi \delta (\cos \epsilon + \tan \delta \sin \epsilon \sin \alpha)$$

$$d\delta = \psi \delta \sin \epsilon \cos \alpha$$

sinusa  $m = \psi \cos \epsilon - \lambda'$

$$n = \psi \sin \epsilon$$

$$d\alpha = -\lambda' \delta + \psi \delta \cos \epsilon + \psi \delta \sin \epsilon \tan \delta \sin \alpha$$

$$d\alpha = \underbrace{(\psi \cos \epsilon - \lambda') \delta}_m + \underbrace{\psi \delta \sin \epsilon \tan \delta \sin \alpha}_{n \delta}$$

$$d\delta = \underbrace{\psi \delta \sin \epsilon \cos \alpha}_{n \delta}$$

$$d\alpha = m\tilde{\tau} + n\tilde{\sigma} \operatorname{tg} \tilde{\sigma} \sin \alpha$$

$$d\tilde{\sigma} = n\tilde{\tau} \cos \alpha$$

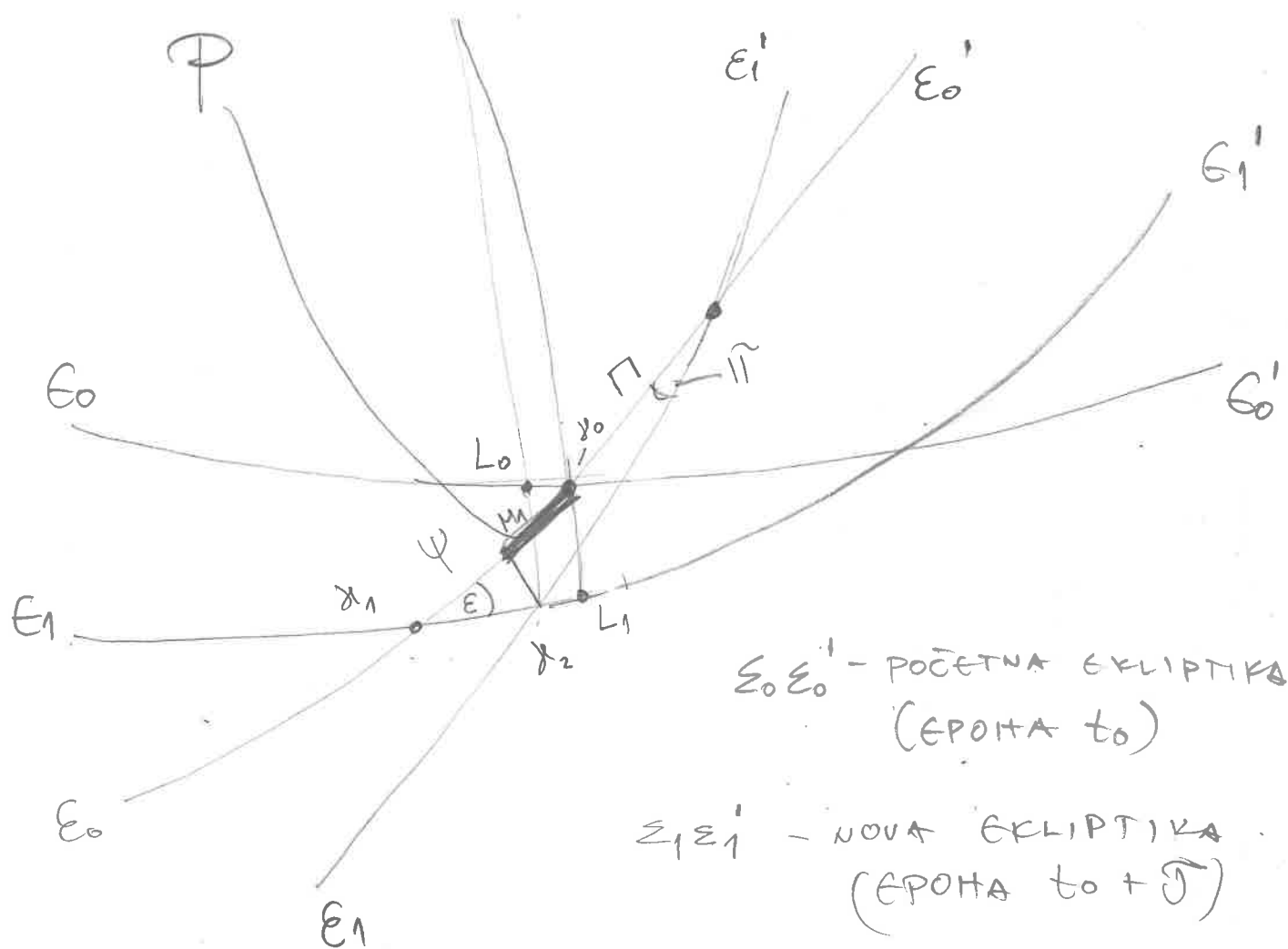
KADA JE  $\tilde{\tau} \neq 1$  god IMAMO

GODIŠNJE PROMENE

$$d\alpha = m + n \operatorname{tg} \tilde{\sigma} \sin \alpha$$

$$d\tilde{\sigma} = n \cos \alpha$$





$$P = \gamma_0 \gamma_1 - M_1 \gamma_1$$

$$P = \psi - \pi' \cos \epsilon$$

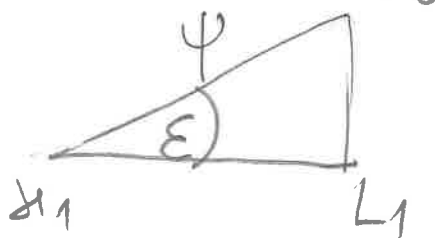
$$P = x_0 x_1 - M_1 x_1 = \psi - \lambda' \cos \epsilon$$

$$m = x_0 L_0 \approx L_1 x_2$$

$$m = L_1 x_1 - x_1 x_2 = \psi \cos \epsilon - \lambda'$$

$$n = x_0 L_1 \approx L_0 x_2 = \psi \sin \epsilon$$

RAVANSKI TROUGAO  $\triangle x_0 x_1 L_1$

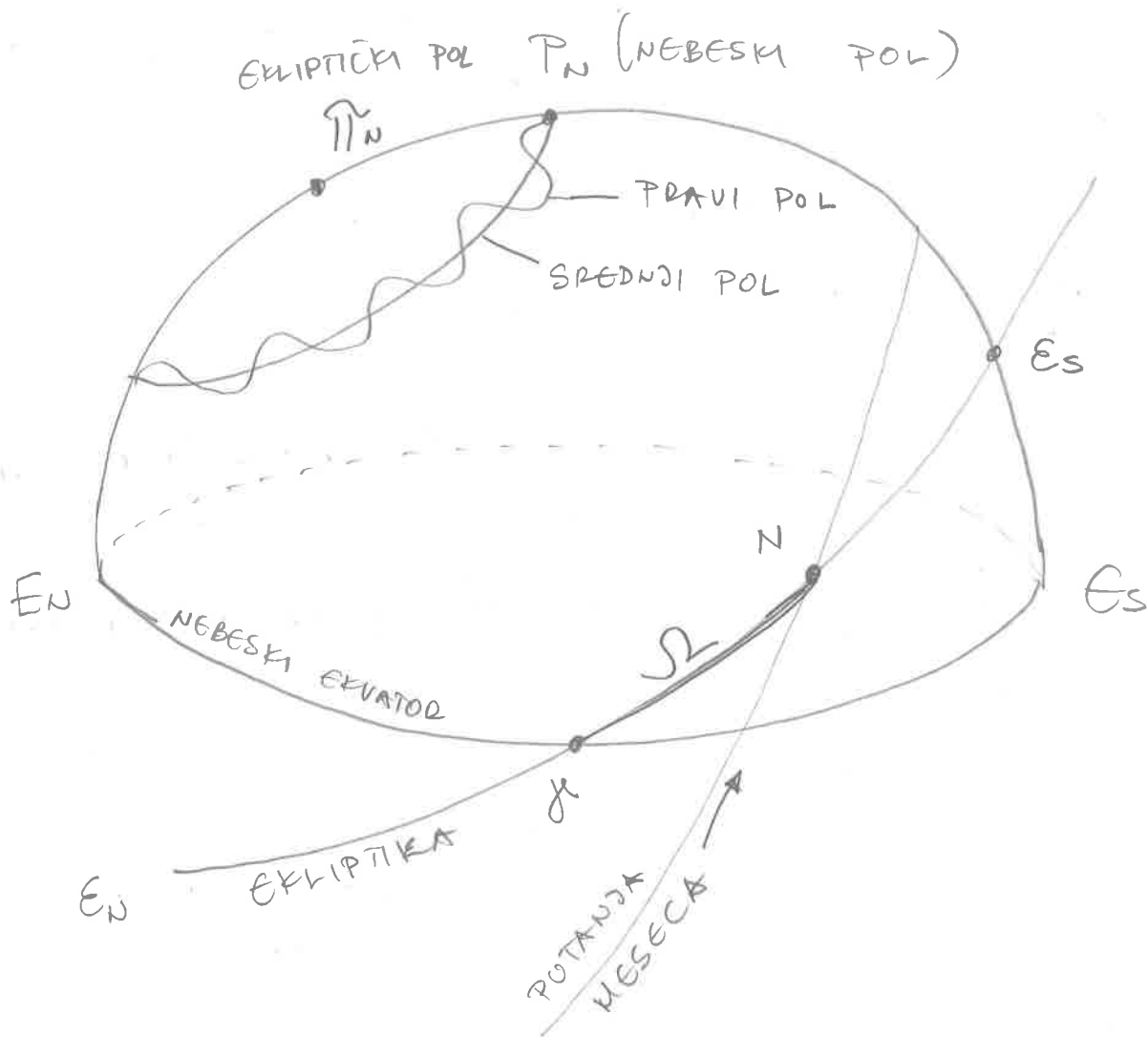


$$x_0 x_1 = \psi$$

$$x_0 L_1 = \psi \sin \epsilon$$

$$x_1 L_1 = \psi \cos \epsilon$$

# ASTRONOMSKA NUTACIJA

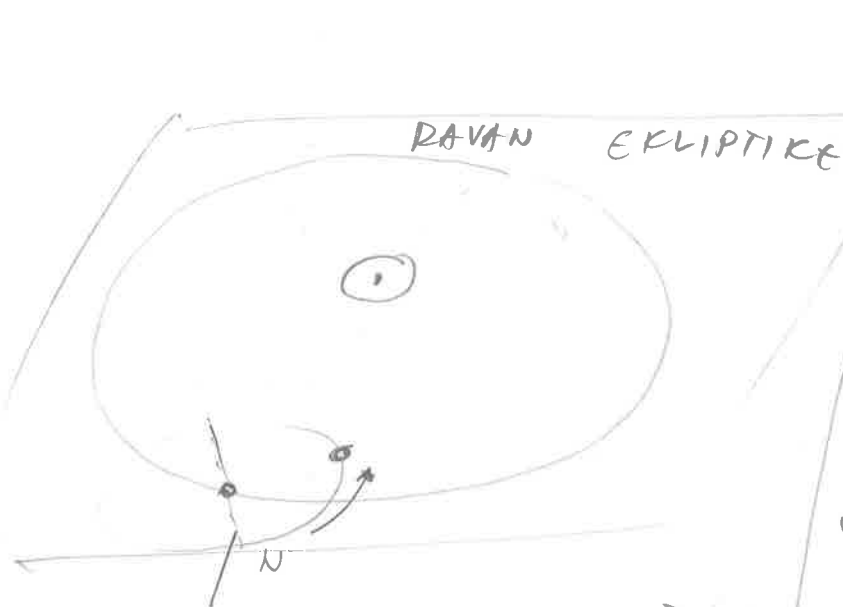


$N$  - VZLAZNI ČVOR MESEČEVE ORBITE

$\Delta\psi$  - PERIODIČNE PROMENE PRECESIJE

$\Delta\epsilon$  - PERIODIČNE PROMENE NAGIBA EKLIPTIKE

- LINIJA ČVOROVA OPIŠE EKLIPTIKU U RETROGRADNOM SMERU ZA OKO 18,6 GODINA, ŠTO PREDSTAVLJA NAJVEĆI NUTACIONI ČLAN.
- NUTACIONI ČLANOVI PREDSTAVLJAJU SE U OBLIKU REDOVA
- ČLANOVI ČIJI JE PERIOD DUŽI OD 15 DANA NAZIVAJU SE DUGOPERIODIČNIM



LINIJA ČVOROVA MASEČEVE ORBITE

NAJVEĆI NUTACIONI ČLAN

NUTACIJA U LONGITUDI

$$\Delta \psi = -17''1996 \sin \Omega$$

NUTACIJA U NAGIBU

$$\Delta \epsilon = 9''2025 \cos \Omega$$

$\Omega$  - LONGITUDA UZLAZNOG ČVORA

# UTICAJ NOTACIJE NA EKVATORSKE KOORDINATE

$\Delta\psi$  - NOTACIJA U LONGITUDI

$\Delta\epsilon$  - NOTACIJA U LATITUDI

UTICAJ LUNI-SOLARNE PRECESIJE

$$\left. \begin{aligned} d\alpha &= \psi\dot{\sigma} (\cos\epsilon + \sin\epsilon \sin\alpha \tan\delta) \\ d\delta &= \psi\dot{\sigma} \sin\epsilon \cos\alpha \end{aligned} \right\} \begin{array}{l} \text{POGLEDATI} \\ \text{KOD} \\ \text{PRECESIJE} \end{array}$$

UTICAJ NOTACIJE SE PREDSTAVLJA KAO

$$\begin{aligned} d\alpha &= (d\alpha)_{\Delta\psi} + (d\alpha)_{\Delta\epsilon} \\ d\delta &= (d\delta)_{\Delta\psi} + (d\delta)_{\Delta\epsilon} \end{aligned}$$

USLED NOTACIJE  
U LONGITUDI

USLED NOTACIJE  
U NAGIBU

UTICAJ NOTACIJE U LONGITUDI (ISTI KAO KOD PRECESIJE)

$$(d\alpha)_{\Delta\psi} = \Delta\psi (\cos\epsilon + \sin\epsilon \sin\alpha \tan\delta)$$

$$(d\delta)_{\Delta\psi} = \Delta\psi \sin\epsilon \cos\alpha$$



## SINUSNO - KOSINUSNI OBRAZAC

$$\sin(90-\delta)\cos(90+\alpha) = \cos(90-\beta)\sin \epsilon -$$

$$\sin(90-\beta)\cos \epsilon \cos(90-\lambda)$$

$$\cos \delta \sin \alpha = -\sin \beta \sin \epsilon + \cos \beta \cos \epsilon \sin \lambda \quad (2)$$

UPOREĐIVANJEM (1) i (2)  $\Rightarrow$

$$\cancel{\cos \delta} d\delta = \cancel{\cos \delta} \sin \alpha d\epsilon$$

$$\boxed{d\delta = \sin \alpha d\epsilon} \quad (3)$$

## SINOSNI OBRAZAC

$$\sin(90-\delta) \sin(90+\alpha) = \sin(90-\beta) \sin(90-\lambda)$$

$$\cos \delta \cos \alpha = \cos \beta \cos \lambda \quad / \text{ DIFFERENCIRAMO}$$

PROVODIVE

$$-\sin \delta \cos \alpha d\delta - \cos \delta \sin \alpha d\alpha = 0$$

$$\cos \delta \sin \alpha d\alpha = -\sin \delta \cos \alpha \underbrace{(d\delta)}_{\substack{\uparrow \\ (3)}}$$

$$\cos \delta \sin \alpha d\alpha = -\sin \delta \cos \alpha \sin \alpha d\epsilon$$

$$\boxed{d\alpha = -\operatorname{tg} \delta \cos \alpha d\epsilon} \quad (4)$$

UKUPNI UTICAJ NUTACIJE

$$\begin{aligned} d\alpha &= \Delta\psi (\cos \epsilon + \sin \epsilon \sin \alpha \operatorname{tg} \delta) - \operatorname{tg} \delta \cos \alpha \Delta\epsilon \\ d\delta &= \Delta\psi \sin \epsilon \cos \alpha + \sin \alpha \Delta\epsilon \end{aligned}$$

GDE SU NAJVEĆI ČLANOVI NUTACIJE

$$\Delta\psi = -17'',2 \sin \Omega \quad \text{NUTACIJA U LONGITUDI}$$

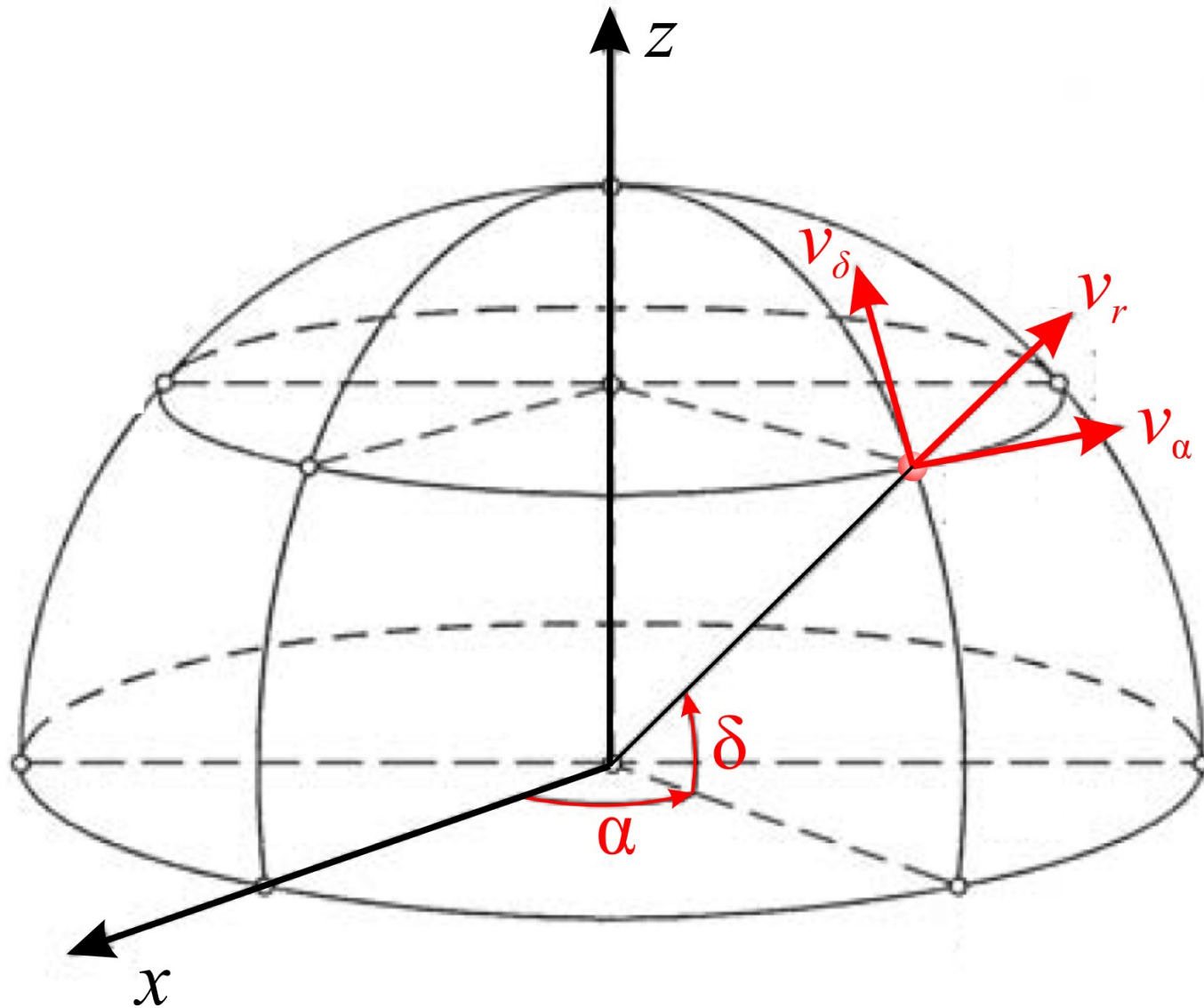
$$\Delta\epsilon = 9'',2 \cos \Omega \quad \text{NUTACIJA U NAGIBU}$$



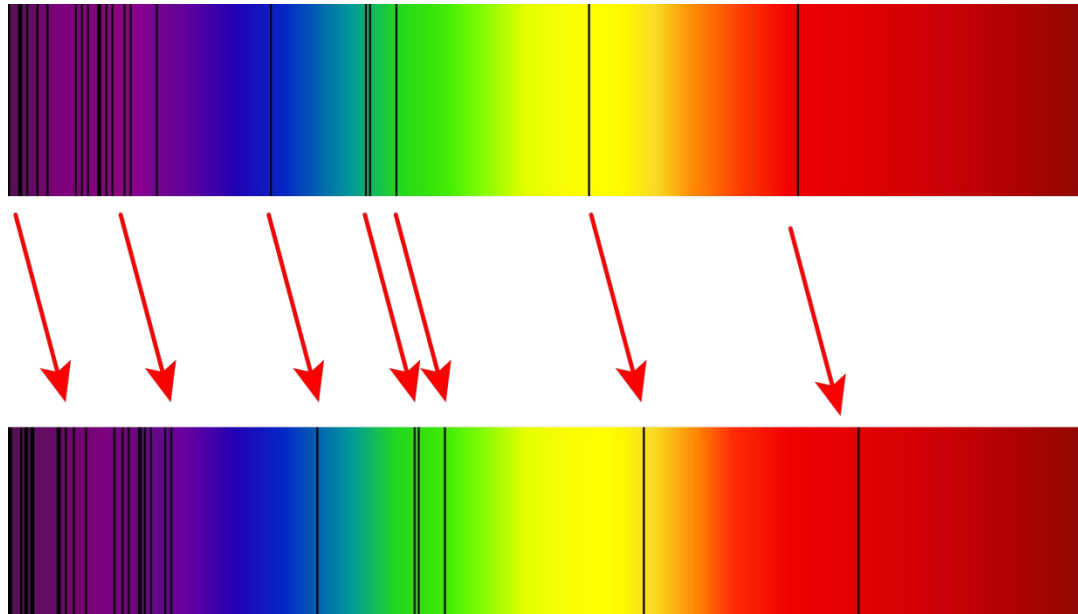
**Sopstveno kretanje**

# Kretanje zvezda

## Od čega zavisi brzina pomeranja zvezde po nebeskoj sferi?



# Radijalne brzine



Doplerov efekt

$$\frac{\Delta\lambda}{\lambda} = \frac{\lambda' - \lambda}{\lambda} = z$$

Klasičan slučaj

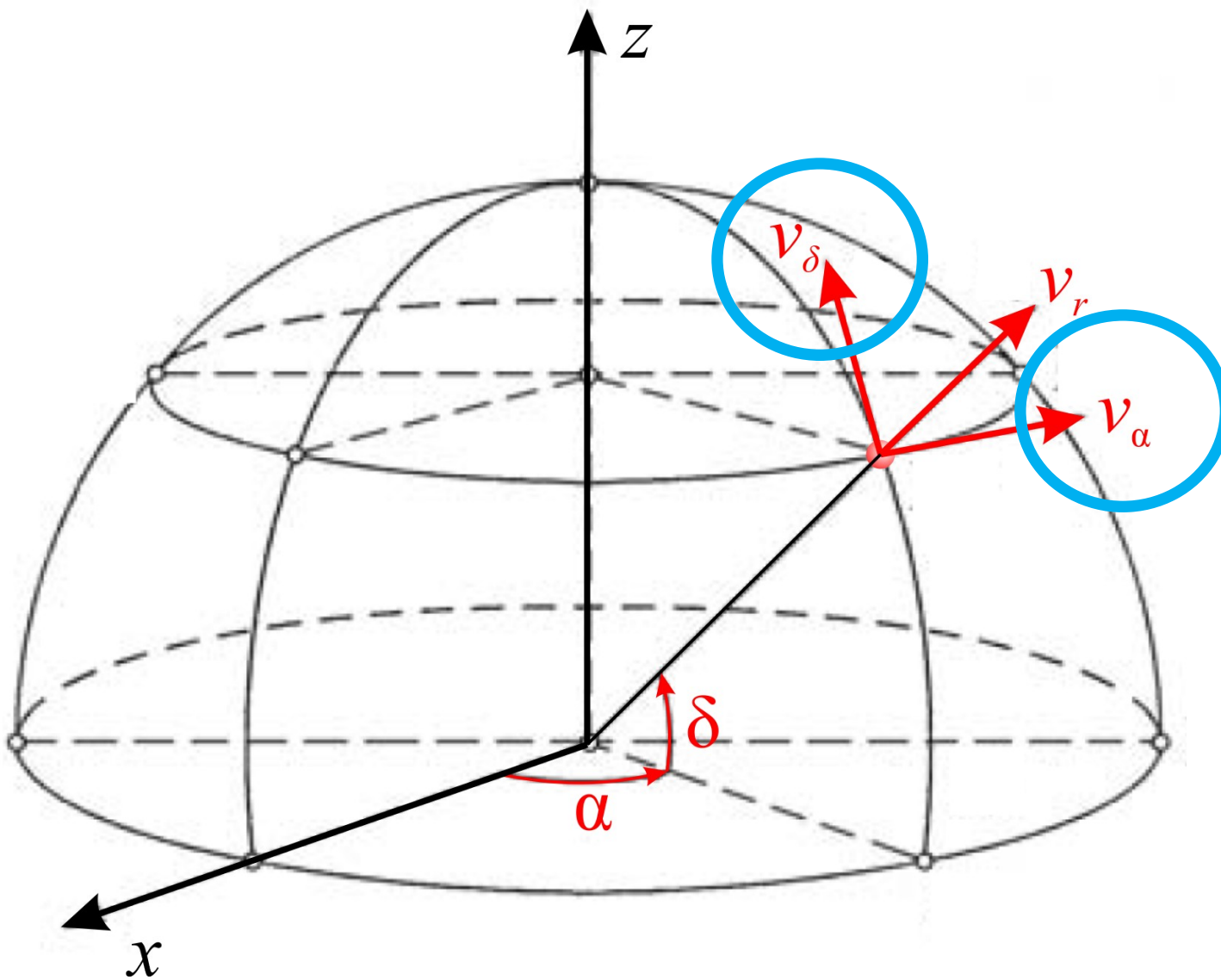
$$v_r = c \frac{\Delta\lambda}{\lambda}$$

Relativistički slučaj

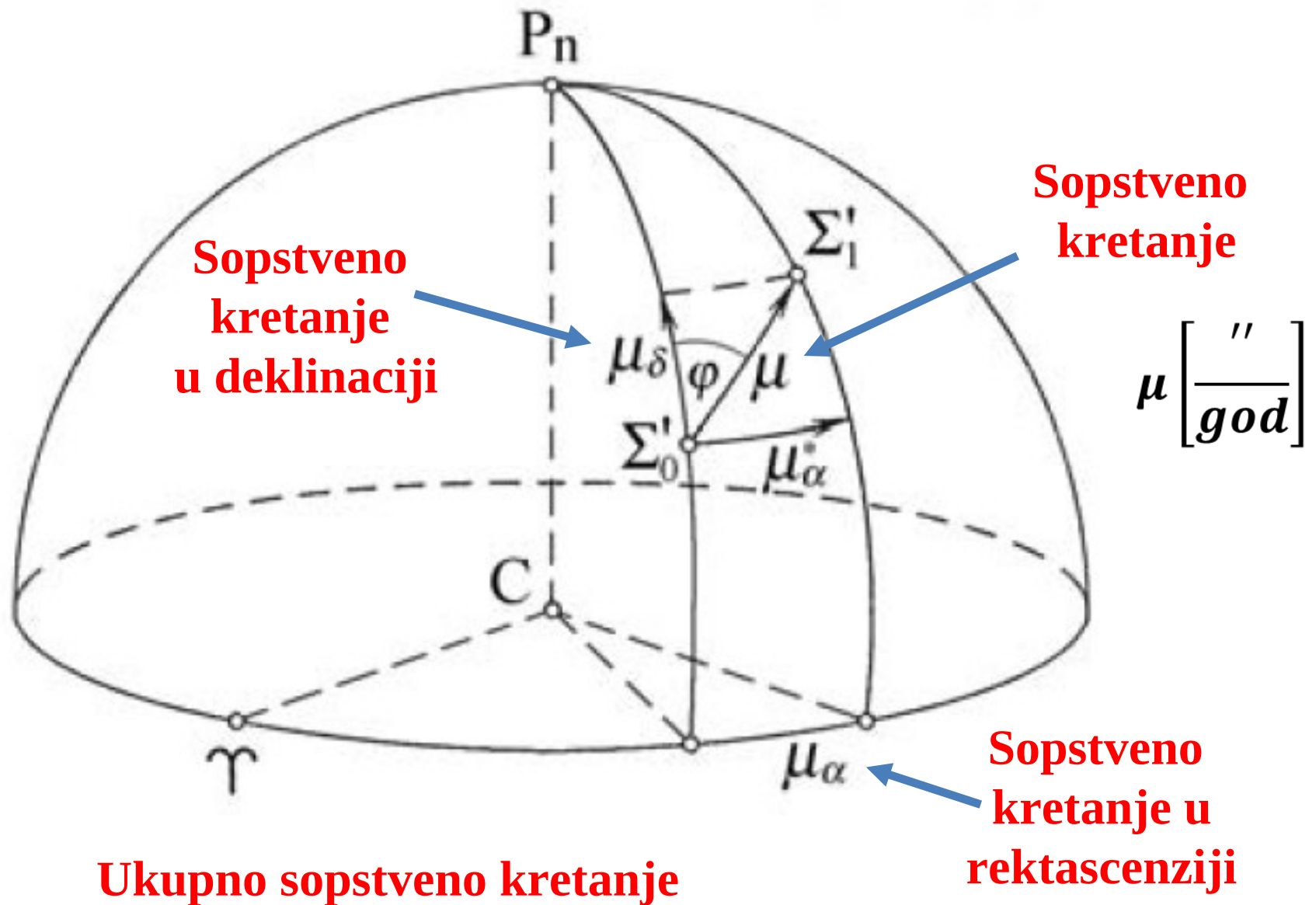
$$\beta = \frac{(1+z)^2 - 1}{(1+z)^2 + 1}$$

$$v_r = \beta c$$

# Tangencijalne brzine



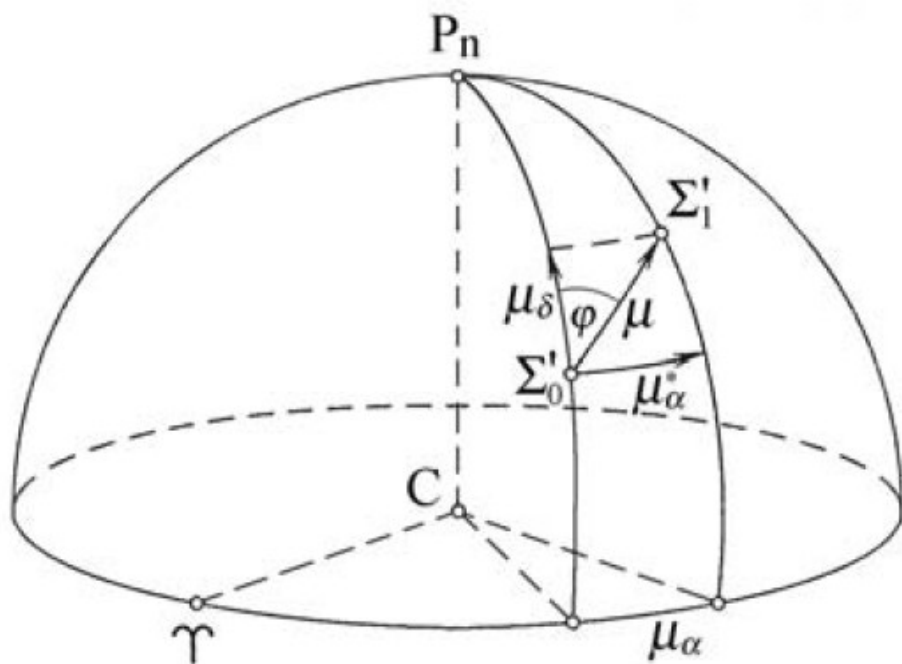
# Sopstveno kretanje



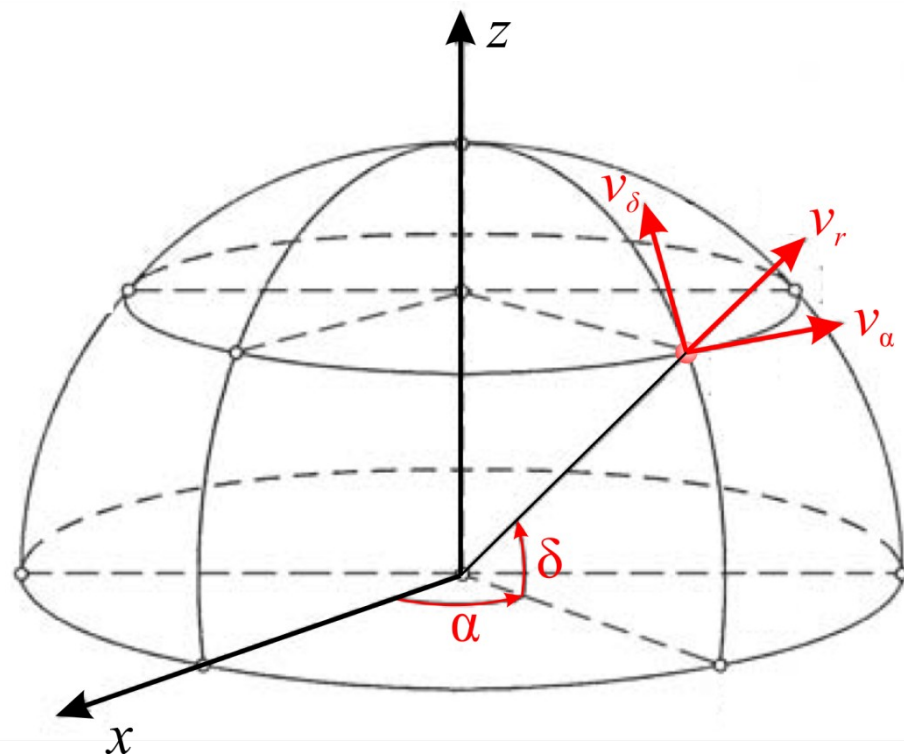
$$\mu^2 = \mu_\delta^2 + \mu_\alpha^{*2} = \mu_\delta^2 + (\mu_\alpha \cos \delta)^2$$

# Sopstveno kretanje

Kako su povezani sopstveno kretanje i tangencijalne brzine?



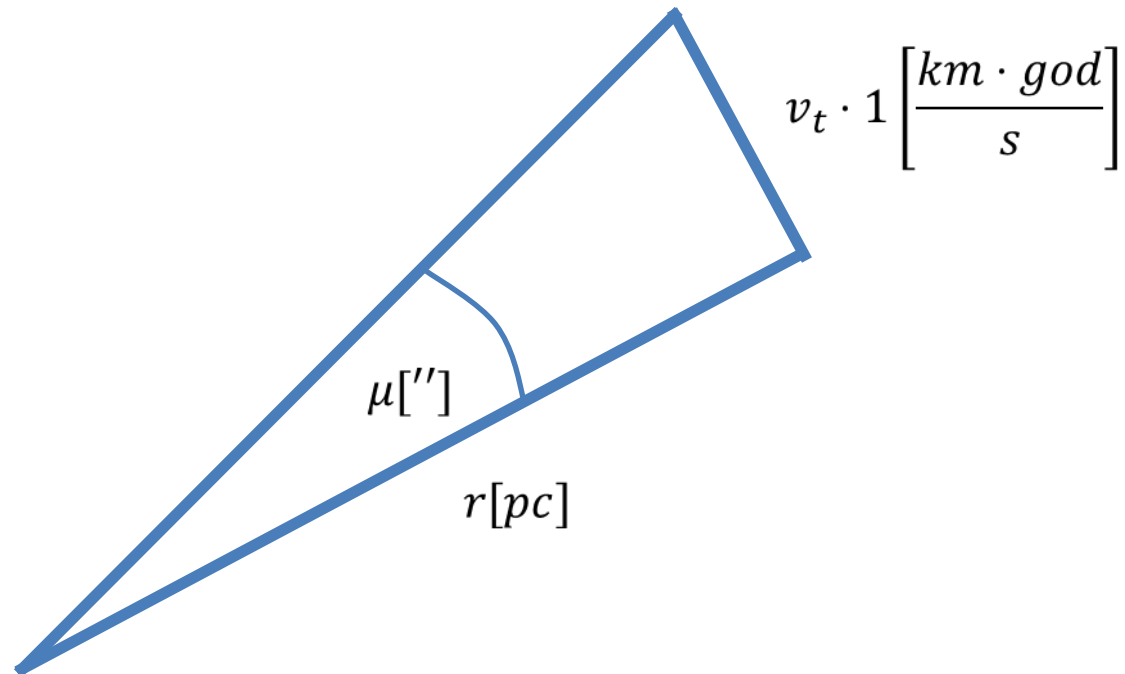
$$\mu \left[ \frac{''}{god} \right]$$



$$v \left[ \frac{km}{s} \right]$$

# Sopstveno kretanje

Kako su povezani sopstveno kretanje i tangencijalne brzine?

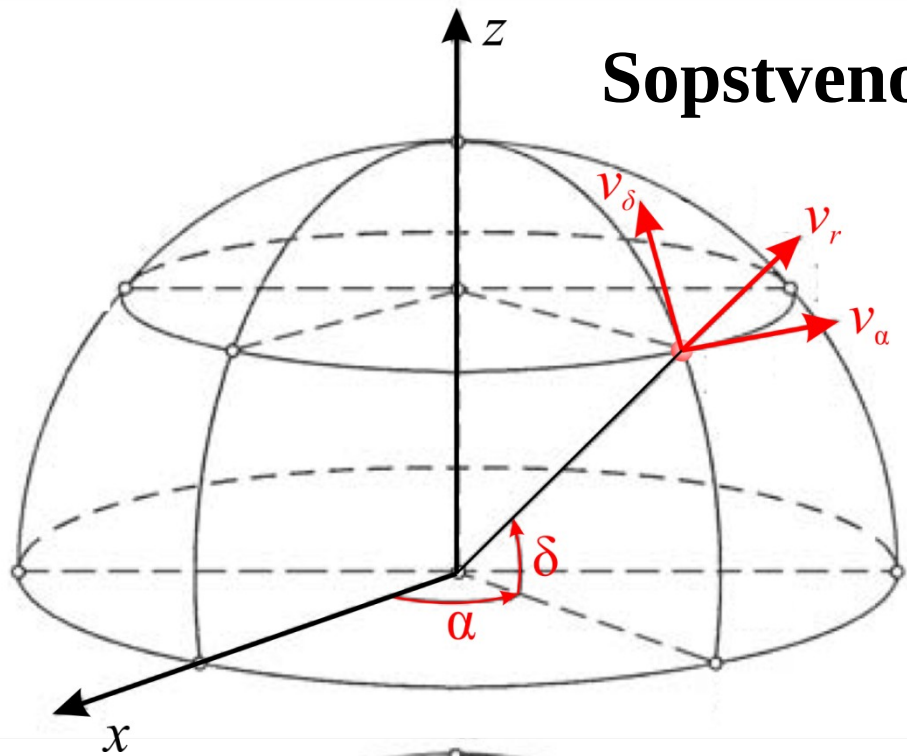


$$v_t \cdot 1 \left[ \frac{km \cdot god}{s} \right] = r[pc] \cdot \mu['] = r\mu[AJ]$$

$$v_t \left[ \frac{km}{s} \right] = r\mu \left[ \frac{AJ}{god} \right] = \frac{\mu}{\pi} \frac{149600000}{31557600} \left[ \frac{km}{s} \right]$$

$$v_t = 4.74 \frac{\mu}{\pi}$$

# Sopstveno kretanje

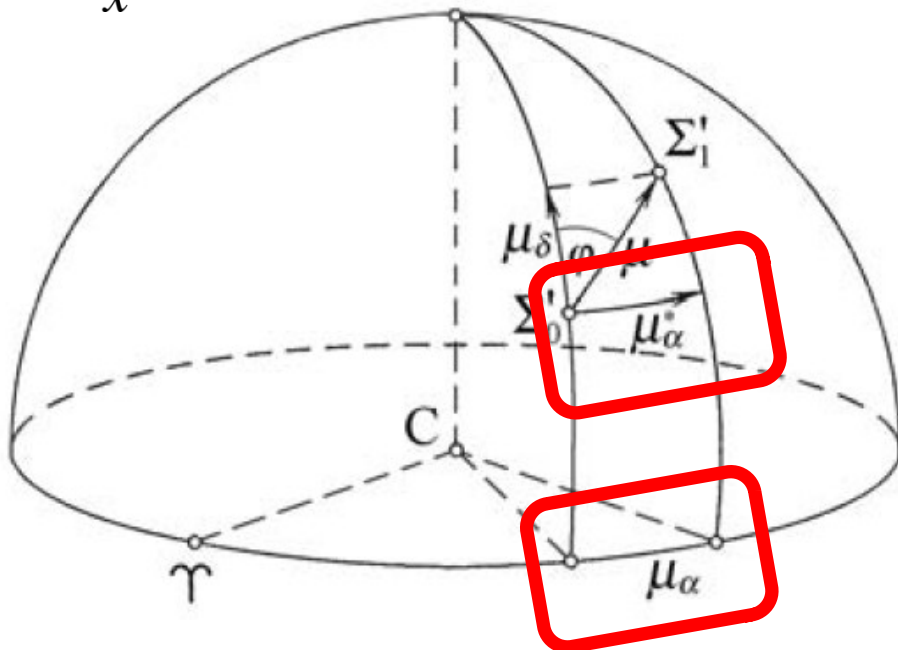


**Barnardova  
zvezda!**

$$v_{\delta} = 4.74 \frac{\mu_{\delta}}{\pi}$$

?

$$v_{\alpha} = 4.74 \frac{\mu_{\alpha} \cdot \cos \delta}{\pi}$$



$$\mu_{\alpha}^* = \mu_{\alpha} \cos \delta$$